Journal of Financial Economics 176 (2026) 104218

Contents lists available at ScienceDirect

Journal of Financial Economics

journal homepage: www.elsevier.com/locate/finec

Social media as a bank run catalyst☆

J. Anthony Cookson [a,*], Corbin Fox [b], Javier Gil-Bazo [c], Juan F. Imbet [d], Christoph Schiller [e]

[a] University of Colorado Boulder, Leeds School of Business, 995 Regent Drive, Boulder, 80309, CO, USA
[b] Clemson University, Powers College of Business, 225 Walter T. Cox Blvd., Clemson, 29634, SC, USA
[c] Universitat Pompeu Fabra; Barcelona School of Economics; UPF Barcelona School of Management, Ramon Trias Fargas, 25-27, Barcelona, 08005, Spain
[d] Paris Dauphine University, Of. P606 Université Paris Dauphine Pl. du Maréchal de Lattre de Tassigny, Paris, 75016, France
[e] The Ohio State University, Fisher College of Business, 2100 Neil Avenue, Columbus, 43210, OH, USA

ARTICLE INFO

Dataset link: Social Media as a Bank Run Catalyst (Original data)

JEL classification:
G21
G28
G14
D83

Keywords:
Bank runs
Social media
Social finance
FinTech

ABSTRACT

After the run on Silicon Valley Bank (SVB) in March 2023, U.S. regional banks entered a period of significant distress. We quantify social media's role in this distress using comprehensive Twitter data. During the SVB run period, banks with high *pre-existing exposure* to Twitter lost 4.3 percentage points more stock market value. Moreover, Twitter pre-exposure interacts significantly with classical run risks to predict greater run severity and greater deposit outflows during Q1-2023, effects unexplained by other banking or market characteristics. At the hourly frequency during the run, high Twitter attention over the past four hours predicts stock market losses, especially for banks with high run risks. By contrast, we find that negative Twitter sentiment does not amplify bank run risks. Rather, our evidence points to a distinctive role of Twitter attention, particularly when tweets are retweeted broadly.

1. Introduction

This paper investigates social media's role in the bank run on Silicon Valley Bank (SVB) and the distress of regional banks in the wake of SVB's failure on March 10, 2023. SVB's failure was preceded by a flurry of Twitter activity by apparent depositors who openly used words like "withdraw" in their tweets. The openness and speed of communication around a bank run was unprecedented, which led to immediate concern about contagion to other banks. Indeed, FDIC Chairman Gruenberg cited social media's potential contagion effects in his May 2023 Congressional remarks (Gruenberg, 2023). Despite this narrative and policy response emphasizing social media's importance to

☆ Philipp Schnabl was the editor for this article. We thank Asaf Bernstein, Jamie Brown, Erica Jiang, Will Mullins, Manju Puri, Klaus Schaeck, Christophe Spaenjers, Brian Waters, Jaime Zender, Philipp Schnabl (the editor), and two anonymous referees for helpful comments. This draft has benefited from conference participant feedback at NBER SI Risks of Financial Institutions, FDIC Bank Research Conference, Community Banking Research Conference, Wharton Conference on Liquidity and Financial Fragility, Finance Organizations and Markets (FOM) Conference, Washington University Finance Conference, Boston Fed Stress Testing Conference, TCU Finance Conference, Cleveland Fed Financial Stability Conference, CEMLA Dallas Fed Financial Stability Conference, Journadas de Economia Industrial Conference, Banco de Mexico Financial Stability Conference, ASU-UA Junior Finance Conference, CEPR/SAFE Conference on Challenges for Financial Intermediaries and Their Supervisors, CEBRA — Fed Board Conference on Nontraditional Data, Machine Learning and National Language Processing in Macroeconomics, and IESE VI Banking Initiative Conference and from seminar feedback at The Ohio State University, Deutsche Bundesbank, the European Central Bank, the Financial Stability Board, the Treasury/OCC, University of Colorado, Pompeu Fabra University, the University of Luxembourg, CEMFI and the Single Resolution Board. Cookson gratefully acknowledges financial support provided by the Michael A. Klump Professorship and the Leeds School of Business. Gil-Bazo acknowledges financial support from the Severo Ochoa Programme for Centres of Excellence in R&D (Barcelona School of Economics CEX2024-001476-S), funded by MCIN/AEI/10.13039/501100011033, and the Spanish Ministry of Science and Innovation (research grant PID2023-153288NB-I00). All remaining errors are our own.

* Corresponding author.
E-mail addresses: tony.cookson@colorado.edu (J.A. Cookson), cfox5@clemson.edu (C. Fox), javier.gil-bazo@upf.edu (J. Gil-Bazo), juan.imbet@dauphine.psl.eu (J.F. Imbet), schiller.96@osu.edu (C. Schiller).

https://doi.org/10.1016/j.jfineco.2025.104218
Received 23 June 2023; Received in revised form 10 November 2025; Accepted 27 November 2025
Available online 1 December 2025
0304-405X/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

this episode, mainstream accounts of the Banking Stress of 2023 offered only incidental mention of social media (e.g., see Acharya et al., 2023).

Prominent perspectives on communication and bank stability offer conflicting predictions on social media's likely role in banking stability (Calomiris and Kahn, 1991; Angeletos and Werning, 2006). On one hand, the potential for bank runs on insolvent banks can achieve discipline (Diamond and Rajan, 2000). According to this view, faster recognition of insolvency via social media would not result in additional risk. On the other hand, a bank run can be modeled as the inefficient equilibrium in coordination models in which communication among depositors plays a central role (Goldstein and Pauzner, 2005). In theory, such coordination plays an important role in modeling insolvency runs (Jiang et al., 2024a), but whether communication on social media amplifies these run risks is an open question.[1]

In this paper, we examine this question using Twitter data on U.S. bank stocks to measure *ex ante* Twitter exposure and *in-run* Twitter attention. We find that greater *ex ante* Twitter exposure predicted larger run period stock losses for banks with high uninsured deposits and large mark-to-market losses, consistent with the idea that social media amplified traditional bank run risks. These conclusions hold using variation in pre-exposure from before the monetary tightening cycle and in specifications that control for a broad set of other bank and market characteristics that could jointly drive run period losses and social media attention. For banks with high run risk, *in-run* Twitter attention predicts larger stock losses at the hourly frequency. These in-run stock losses are larger in hours after highly-retweeted tweets, pointing to a distinctive role of social media.

We now describe our analysis in greater detail. We measure Twitter pre-exposure using the logged number of tweets about each bank between January 1, 2023 and February 15, 2023, nearly one month before the SVB run. We validate that Twitter pre-exposure is driven by optimism rather than skepticism about bank stability and is largely unrelated to traditional run risk factors.[2] Because investor social media users tend to be fanatics (Pedersen, 2022; Cookson et al., 2023), pre-run Twitter *exposure* strongly predicts in-run Twitter *attention*.[3] Our initial tests relate Twitter pre-exposure to run-period stock losses in cross-sectional regressions. Unconditionally, we find that a standard deviation increase in Twitter pre-exposure predicts a 4.3 percentage points larger stock market loss during the SVB run period. This estimate is similar to the decline in run period bank stock prices associated with a standard deviation increase in uninsured deposits, and it is also robust to controlling for uninsured deposits. Consistent with the model intuition in Jiang et al. (2024a), we find that the interaction between mark-to-market losses and the percentage of uninsured deposits — which we dub 'balance sheet risk' or 'bank run risk' — matters disproportionately for the decline in bank stock prices during the run.

Related to the idea that social media amplifies run risks, we find that Twitter pre-exposure *interacts* significantly with bank run risk to predict larger run-period bank stock losses. The significant interaction between Twitter pre-exposure and run risk is robust to including controls for bank size and other natural drivers of attention to bank stocks, such as traditional media coverage in Q4-2022, analyst coverage in Q4-2022, stock returns in the final two quarters of 2022, returns around 2022 FOMC announcements, retail attention, and a host of bank stability and funding characteristics.[4] Further, the findings are similar if we measure *ex ante* Twitter exposure using alternative pre-periods, dating back to October 2022. We also obtain similar findings after dropping large banks that experienced a "flight to safety" during the run period (Caglio et al., 2023), and our findings are robust to dropping SVB itself from the sample.

To mitigate concerns that run-period stock losses do not proxy well for run severity, we complement the analysis of run-period returns with an analogous cross-sectional specification but with percentage of Q1-2023 deposit outflows as the dependent variable. We find that high Twitter pre-exposure predicts more deposit outflows, especially of uninsured deposits. These outflows are particularly strong for banks with a high initial fraction of uninsured deposits and a large fraction of mark-to-market losses. Although the deposit data are at the quarterly frequency, the consistency of this evidence with our main tests helps to validate our approach to proxy for run severity using run period stock market losses. As with the main specifications, these conclusions are robust to including a battery of controls.

We next analyze the content of the conversation surrounding the run by examining the content of tweets, as well as who authored them. The most notable banks with Twitter pre-exposure are also banks with more intensive "run" discussions during the run period (e.g., tweets that mention "withdraw"). For example, from March 8 through March 14, users posted 6528 'run' tweets about SVB, which is roughly five times the next most run discussed ticker (i.e., First Republic Bank, FRC, which also faced severe distress and failed on May 1, 2023). We see a similar pattern with tweets that mention contagion: SVB had 9662 such tweets during the run period. However, the mechanism linking Twitter pre-exposure to run-based tweets and bank run price declines is not limited to SVB. In a cross-sectional regression, we find that controlling for in-the-run discussion of "run" and "contagion" significantly reduces the coefficient on Twitter pre-exposure.

We also provide contextual evidence on whose tweets began the run using some unique features of SVB and the identities of depositors in Silicon Valley Bank. This evidence exploits the detailed and high frequency nature of the Twitter data. We note that what matters for run risk is conversation among *depositors*. In the Twitter data, we track conversations about SVB's ticker "SIVB" separately from more general conversations about SVB (i.e., "SVB," "Silicon Valley Bank", etc.) to distinguish investor-contributed tweets from general conversation. As Fig. 1 shows, investor-centric "SIVB" tweets spiked distinctly before mentions of "SVB" or "Silicon Valley Bank." This pattern is consistent with depositors communicating on Twitter about withdrawing their deposits from the bank after investment-specific conversation raised their concern about issues at the bank.

Aside from the timing of investor versus general tweets about SVB, we also employ information on SVB's likely depositors who were

[1] To illustrate why social media can amplify bank run risk, consider the model intuition in Jiang et al. (2024a). They show formally that run risk depends on depositors' beliefs about the fraction of "awake" uninsured depositors s, which is driven partly by expectations of bank health. Jiang et al. (2024a) simulate scenarios where all or half of uninsured deposits run (i.e., $s = 1$ and $s = 0.5$) to help quantify system-wide bank run risk. Since s varies across banks (i.e., s_i), our findings suggest that greater exposure to social media amplifies bank run risk by increasing s_i.

[2] The vast majority of pre-run tweets are never retweeted during the run period. For those that are, their retweets spike after the run begins, reflecting social propagation of information once the run is already underway.

[3] While Pedersen (2022) introduced the term "fanatics" to the literature, a growing empirical literature on investor social media supports the idea that investors are fanatic about stocks they discuss on social media. For concreteness, a fanatic is someone who posts consistently and optimistically about the same stocks over time. This behavior is common on investor social media. For example, investor social media posts are disproportionately bullish: e.g., 80% of self-labeled posts on StockTwits are bullish rather than bearish (Cookson and Niessner, 2020), but similar findings exist on other platforms as well (Dim, 2025; Müller et al., 2023). These users typically post only about one or two stocks, and optimists are very unlikely to change their mind. For example, Cookson et al. (2023) shows that users who were bullish about a particular stock are more than 95% likely to be bullish about that same stock in posts up to 50 days later. In the context of our bank sample, we further validate that users of investor social media are fanatic before the SVB run.

[4] We also verify that our central result is robust in a specification curve analysis, a methodology proposed by Simonsohn et al. (2020), which estimates *all* possible combinations of control variables and interactions with run risk, benchmarked against similar specifications in a permutation test.

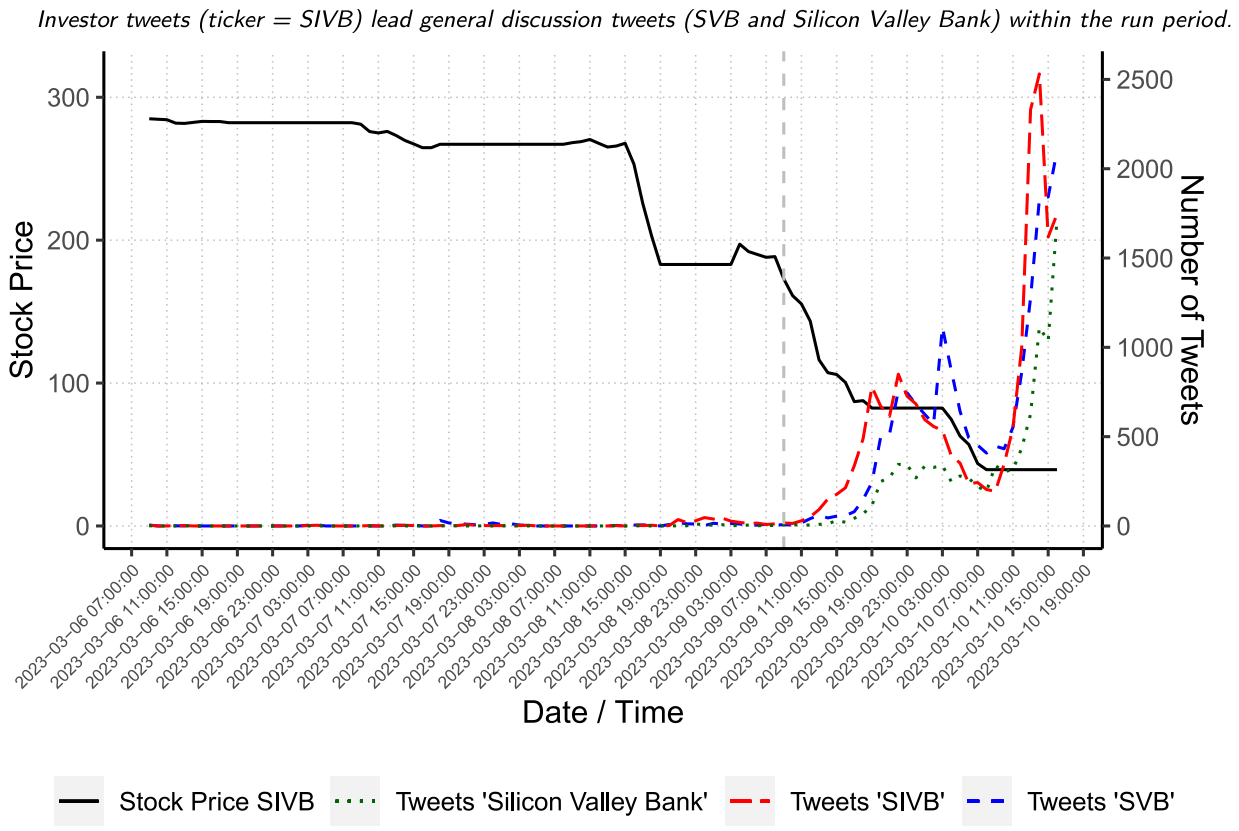


Fig. 1. *Silicon Valley Bank Stock Price and Twitter Mentions.* This figure shows the stock price of Silicon Valley Bank (SIVB) (black solid line), as well as the number of tweets mentioning "SIVB" (blue dashed line), "SVB" (red long-dashed line), and "Silicon Valley Bank" (green dotted line), over the period from March 06, 2023 at 9am (market open) to March 10, 2023 at 4pm (market close). The stock price is indicated on the left *y*-axis and the number of Tweets is indicated on the right *y*-axis. The gray vertical dashed line indicates March 09, 2023 at 9am.

highly concentrated in the 'tech community.' The tech community comprises an active segment on Twitter. We classify tech community users based on founder-related words in user profiles (e.g., "entrepreneur" or "founder"). Consistent with these likely depositors playing a central role in the bank run, tech community tweets spike distinctly after the initial increase in tweet volume, and they are more likely to mention SVB than SIVB (see Fig. 2). These findings about SVB in particular point to the importance of highly concentrated and highly networked communication for bank run risk.

Unlike the main tests that rely on Twitter pre-exposure, a concern with these in-run contextual analyses is that it is unclear whether the run period tweets occur before versus after the bank's stock market losses. To better understand this timing and to provide sharper tests of mechanisms, we analyze *hourly* bank stock returns in a panel setting. These tests also allow us to evaluate which aspect of social media exposure, Twitter attention or sentiment, drives the amplification of bank run risks we see in the Twitter pre-exposure tests. We first confirm that banks with high run risk experienced much larger negative run-period returns than banks with low run risk, consistent with our cross-sectional tests. Crucially, this divergence emerges specifically at the start of the SVB run in the morning of March 9, 2023 in the hourly sample.

Next, we estimate a specification that interacts an indicator for the run period (i.e., post-March 9 at 9 a.m.) with an indicator for high recent Twitter *attention* (i.e., the number of tweets in the past four hours was above the median). We estimate this specification separately for banks with high versus low run risk (i.e., a median split on run risk, which is *% Uninsured* times *% Loss MTM*). This specification distinguishes the timing of Twitter attention from returns since it considers lagged Twitter attention. In the high run risk sample, we find a large, significant negative interaction between lagged Twitter attention and the run-period indicator, but we see no significant interaction in low run risk banks. This finding supports our interpretation that Twitter attention amplified the severity of the run for banks with more run risks. Moreover, the differential impact of Twitter attention by high versus low run risk is similar in tests that exclude SVB from the sample, when we exclude too-big-to-fail banks, and the estimated coefficients are larger when we consider a tighter window around the run on SVB. These conclusions are robust to controlling for traditional news coverage and lagged bank stock returns, as well as including bank fixed effects, which hold constant any time-invariant bank characteristics (i.e., given the sample period in early March, these fixed effects are especially powerful, for example, holding constant factors that vary at the bank-month level). By controlling for lagged bank stock returns and the corresponding interactions, we address concerns that tweets simply reflect past returns.

We present three other tests that shed light onto the nature of social media's amplification of run risks. First, we measure Twitter sentiment separately from Twitter attention. When we include Twitter sentiment in the hourly return specification, we estimate a near zero interaction with the run period indicator in the high run risk subsample. Together with the main finding on Twitter attention, this finding highlights a distinctive role for *attention* in amplifying bank run risks. Second, we measure recent Twitter attention in the past hour and the past two hours, as robustness to our main four-hour window. Our findings are similar irrespective of the lag, indicating that the most recent Twitter activity drives our results. Third, we consider heterogeneity in how often a bank stock's tweets have been retweeted. The impact of Twitter attention is driven mostly by banks and periods for which tweets were retweeted heavily. These findings draw a link between our results and the social nature of social media as evidenced in the intensity of retweets.

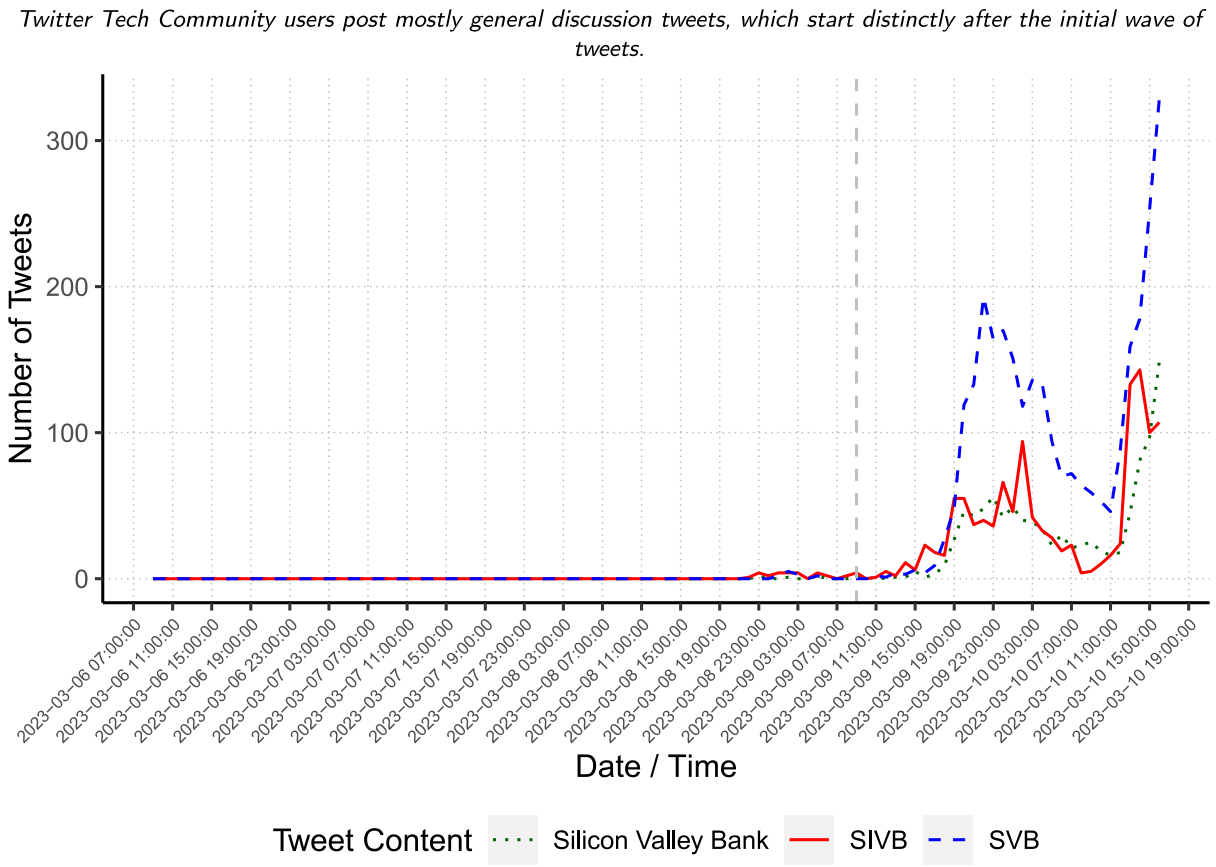


Fig. 2. *Tech Community Tweets about Silicon Valley Bank.* This figure shows the number of tweets mentioning "SIVB" (blue solid line), "SVB" (red dashed line), and "Silicon Valley Bank" (green dotted line), over the period from March 06, 2023 at 9am (market open) to March 10, 2023 at 4pm (market close). The sample includes only tweets posted by users who are part of the tech community as defined by corresponding dictionary of terms in the Twitter user biography. The gray vertical dashed line indicates March 09, 2023 at 9am.

As a final leg of our empirical analysis, we exploit the high frequency timing of tweets by adapting the approach in Bianchi et al. (2023, 2024). These tests help rule out confounding events from outside of a narrow window from just before the tweet to 5 min afterwards, providing sharper identification about which tweets matter for bank stock returns. Using minute-level stock price data from FirstRate, we estimate a significant negative impact of neutral sentiment tweets on bank stock returns posted during the run period, while this effect is small and insignificant prior to the run. Additionally, tweets with more negative sentiment correspond with lower returns, especially during the run period. The impact of negative sentiment is roughly ten times stronger if the tweets are authored by someone in the 'tech community,' or when the tweets contain words in the 'run' or 'contagion' dictionaries. These results are robust to excluding SVB from the sample. However, in these tests, we estimate a near-zero shift in the impact of tweet sentiment for banks with high run risk versus those with low run risk, consistent with our finding at the hourly frequency that Twitter attention does but sentiment does not amplify bank run risks.[5]

Our main contribution is to provide evidence of social media exposure and attention as a bank run risk. Diamond and Dybvig (1983)'s insight is that coordination of run beliefs is foundational to bank run models. Building on this insight, scholars developed models of coordination via communication, which have enjoyed broad application in our understanding of bank runs and financial crises (Peck and Shell, 2003; Angeletos and Werning, 2006; Brunnermeier, 2009; Acemoglu et al., 2015). Since the Global Financial Crisis and the rise of FinTech lenders, research has turned to investigate the risks posed by these new forms of banking (Jiang et al., 2024b; Buchak et al., 2024, 2018; Fuster et al., 2019), and the incentives of run-prone uninsured depositors in generating possible bank run equilibria (Egan et al., 2017; Drechsler et al., 2024; Jiang et al., 2024a). Relative to this literature, which focuses on bank balance sheets and risk taking, our contribution is to highlight how social media as a communication technology can amplify existing risks identified in the literature. The implication that social media exposure matters for banking stability is potentially troubling because social platforms can spread inaccurate information, which could serve as a sunspot that leads to bank runs.[6]

Our research also relates to the banking literature that focuses specifically on financial contagion via social networks (Calomiris and Mason, 1997; Iyer and Puri, 2012). Like our work, this research focuses on communication as a contagion mechanism in bank runs. However, the literature's emphasis has been on how communication spreads through traditional social networks such as immigrant networks or word of mouth (e.g., Kelly and Gráda, 2000). Our research is distinct from this prior research because we provide evidence on the *nature* of run-relevant communication and its relation to bank run risks.

[5] Since these high frequency tests are at the tweet level (one tweet is one observation), they effectively condition on one unit of attention to a bank stock. Thus, the high frequency results are silent about attention's amplification role while corroborating the hourly finding for sentiment.

[6] Other work has emphasized the role of trust in the financial system (Iyer and Puri, 2012; Traweek and Wardlaw, 2025). Indeed, trust is instrumental in encouraging participation in banking and financial markets more broadly (Gurun et al., 2018; Brown et al., 2019; Stein and Yannelis, 2020; Dupont, 2022).

Specifically, our findings emphasize the importance of depositors ('tech community' users), investor-centric versus general tweets ($SIVB versus SVB), conversations about 'running' and 'contagion,' and the role of social propagation via retweets. Our tests also draw an important distinction between social media attention and sentiment. Moreover, on a conceptual level, social *media* is distinct from offline social connections in at least three ways. First, social media posts spread rapidly in comparison to word of mouth communication. Second, information posted to Twitter is visible publicly, which transmits information well beyond close personal connections. Third, widely read tweets can be contributed by *anyone* on the platform, enabling social media to reflect information from many sources. These aspects of social media naturally interact with one another to lead to rapid and widespread sharing of ideas, run intentions, and contagion concerns.

There is a longstanding debate about whether runs are driven by communication about run intentions about otherwise healthy banks (Goldstein and Pauzner, 2005) or the release of information that leads to the unwinding of fundamentally insolvent banks (Calomiris and Mason, 1997). Social media can amplify bank runs through either of these mechanisms. Relating to run intentions, social media serves as a natural and public 'sunspot' (Morris and Shin, 2000), which can coordinate depositor beliefs around the run equilibrium. Relating to the information channel, banks may be troubled due to their limited hedging of interest rate risk, an aspect of banking risk that has received considerable attention (e.g., Flannery, 1981; Flannery and James, 1984; Purnanandam, 2007; Begenau et al., 2015; Drechsler et al., 2021). Rapid communication of these risks on social media could lead to faster and more effective discipline of insolvent banks, which is core to the functioning of banks (e.g., see Gorton and Pennacchi, 1990 Calomiris and Kahn, 1991, Diamond and Rajan, 2000, Pennacchi, 2006 and Granja, 2013).

Our work also relates to a broader literature in economics on communication and contagion in networks (Chwe, 2000; Van Bommel, 2003; Golub and Jackson, 2010; Elliott et al., 2014; Falato et al., 2021). With the advent of myriad technologies to communicate, there has been research into the real effects of this communication, particularly in politics (Parmelee and Bichard, 2011; Müller and Schwarz, 2021). In this respect, our contribution advances what is known about communication technologies such as radio (Strömberg, 2004) and television (Kearney and Levine, 2019). Notably, we closely relate to Ziebarth (2017), which studies the effect of radio on bank distress during the Great Depression. In contrast to radio, which provides sparse information and one-way communication, social media aggregates information from many sources, potentially making it a stronger coordination device in the context of bank runs. As a communication technology, social media complements recent technological changes in banking, such as digital banking (Haendler, 2022), which have been linked to destabilizing bank deposits (Koont et al., 2023).[7]

We also contribute to the social economics literature (Levy, 2021; Chetty et al., 2022a,b). In particular, we complement recent work that has emphasized social transmission of ideas and peer effects (Akçay and Hirshleifer, 2021; Hirshleifer, 2020; Kuchler and Stroebel, 2021; Cramer and Koont, 2021).[8] Bank runs are by their nature a social phenomenon, and social media is a natural place to study the formation and transmission of ideas. Much of this literature has studied investment outcomes (Han et al., 2022; Pedersen, 2022; Cookson et al., 2023), while only a few studies have examined real outcomes (e.g., Hirshleifer and Plotkin, 2021; Cookson et al., 2025). Although some research on social media has investigated political outcomes and ideologies (Müller and Schwarz, 2023), we are the first to provide direct evidence of a social transmission channel via social media for bank runs.

2. Data and measurement

2.1. Twitter sample

Using academic access to Twitter's API, we collect all original tweets from 1 January 2020 until 14 March 2023 containing at least one cashtag (i.e., $ followed by the company's ticker) for all depository institutions with the three-digit SIC code 602, 603, or 609. The practice of using cashtags on Twitter began in 2012, and it is now commonplace on investor social media to reference publicly traded companies using their cashtag. For our main sample, we focus on original tweets (i.e., no retweets) that are written in English. These sample filters yield a sample of 5,399,740 original tweets about banking firms from 2020 until March 2023. We complement our main sample with a separate sample of all retweets with cashtags from 1 January 2023 until 31 March 2023. Using similar filters, this sample consists of 765,224 retweets. Tweet-level variables include the text of the tweet, a timestamp, an author identifier, and number of retweets.

In addition, we separately query author-level data for each of the 544,888 unique Twitter users who authored at least one tweet in our sample. Author-level variables provide information about the authors' accounts including number of followers, date of account creation, and a user-provided description.

For hourly tests and cross-sectional tests, we aggregate the number of messages (of different types) to the appropriate level, but for the high-frequency analysis, we use the precise timestamp.

2.1.1. Processing and scoring of tweets

We score the sentiment of each tweet with Valence Aware Dictionary and sEntiment Reasoner (VADER), a sentiment classifier designed specifically for social media. Hutto and Gilbert (2014) propose VADER as a sentiment classifier for short messages that contain social text, like tweets, and show that it outperforms naive dictionary-based methods. We use the VADER algorithm to compute a raw sentiment score based on the words contained in the tweet. The VADER algorithm is an aggregate of a negative component, a neutral component, and a positive component. To compute sentiment, the VADER algorithm aggregates across these sub-components to form a sentiment score for each tweet, which is a number between −1 (very negative) and 1 (very positive).

For tweet level tests, we employ the negative and positive VADER components, separately. We do this to capture asymmetry in the impact of negative content versus positive content. In the context of sentiment analysis in accounting and finance, it is common to include positive and negative keywords separately (Loughran and McDonald, 2011, 2020).

2.1.2. Content dictionaries

We define three content dictionaries of terms used in tweets about banks during this period: "balance sheet", "run behavior", and "contagion." We construct these dictionaries by adapting an iterative method in the spirit of Gentzkow and Shapiro (2011), similar to the approach used by Cookson et al. (2020) for investor tweets.

For each topic, we identify a small set of seed words. For example, the balance sheet dictionary includes 'hold-to-maturity' or 'HTM'; For run behavior, 'run' and 'get out'; For Contagion, 'systemic' or 'spillover'; The full set of seed words is italicized in the contextual dictionaries in Table 1. Using these seed words, we identify the subsample of tweets containing those words, which we use to compute the frequency distribution of words used. We identify the top 40 most *salient* words by topic

[7] Our evidence on social media complements other research on the 2023 regional banking crisis, which revealed issues with commercial real estate lending (Jiang et al., 2025), hold-to-maturity accounting and interest rate hedging (Granja et al., 2024). Others emphasize flights to safety (Caglio et al., 2023), savings gluts (Vuillemey, 2023), the importance of interest rate hikes (Haddad et al., 2023), and the stability effects of banking on uninsured deposits (Drechsler et al., 2024).

[8] In a narrow sense, we relate to literature on financial social media (Chen et al., 2014; Giannini et al., 2019; Cookson and Niessner, 2020; Cookson et al., 2024a; Gil-Bazo and Imbet, 2025). These papers focus on asset market effects (Renault, 2017; Cookson et al., 2020; Bianchi et al., 2023, 2024), whereas our focus is on the broader conversation that emerged about depositor issues.

Table 1
Contextual dictionaries for classifying types of tweets.

Balance sheet	Run behavior	Contagion	Tech community
duration	*run*	*systemic*	VC
cover & cash	*withdraw*	*spillover*	entrepreneur
mortgage backed securities	deposit money	fed	start up
mismatch	access accounts	regulator	startup
long maturity	pull & out	#contagion	founder
maturity mismatch	*get & out*	backstop	_venture
marked to market		whole system	venture capital
mark to market		spreading	
portfolio management		sparks	
liquidity		broader effects	
insured deposits		financial system	
MBS		meltdown	
hold to maturity		*contagion*	
HTM			
portfolio of loans			
liquidity management			
uninsured deposits			
balance sheet			

This table presents the keyword dictionaries for the major content dictionaries that we employ to classify tweets in to "Run," "Contagion,", "Balance Sheet," as well as "Tech Community" authors. For each of our content dictionaries we provide the seed words used, in the iterative creation process, in italics.

in comparison to the overall distribution of words using the saliency measure from Goldsmith-Pinkham et al. (2016). We then consolidate words into meaningful bigrams that emerged from the salience analysis (e.g., "liquidity" and "management" → "liquidity management"), and eliminate terms that appear in multiple dictionaries at the same time (e.g., 'bank' appeared in all dictionaries). The resulting contextual dictionaries, summarized in the first three columns of Table 1, flag tweets that cover specific topics, which helps identify mechanisms.

In addition, we also construct a user-level dictionary by text mining the user description field for whether the user is part of the "tech community", which is an important subset of uninsured depositors in SVB who actively post on Twitter. To do this, we employ a list of terms around the idea of "founder", "entrepreneur", and more generally, the VC-backed startup industry. This dictionary is reported in the final column of Table 1. Tweets flagged as "tech community" tweets are those authored by a user with at least one of these terms in their user profile description.

2.2. Bank balance sheet information

To measure balance sheet risk or 'run risk', we construct two key measures: *% Uninsured* and *% Loss MTM*. To calculate the percentage of uninsured deposits, i.e., *% Uninsured*, we collect data on the total value of bank deposits below (data field RCONF049) and above (data field RCONF051) the FDIC insurance threshold of $250,000 from Call Report Schedule RC-O for Q4 of 2022. We then calculate the *% Uninsured* as RCONF051/(RCONF049+RCONF051). *% Uninsured* is an important determinant of run risk. For example, Drechsler et al. (2024) argue that it is the uninsured deposit franchise (not the entire deposit franchise) that makes a bank more vulnerable to runs. For this reason, *% Uninsured* predicts run exposure.[9]

Second, for *% Loss MTM*, we construct a market-based measure similar to Jiang et al. (2024a). We use banks' asset holdings as reported at the start of Q1-2022 and changes in Treasury Bond indexes throughout 2022 to estimate market implied asset value losses by the end of 2022. To estimate *% Loss MTM*, we first collect the value of residential mortgage backed securities (RMBS) and non-residential MBS, loans secured by family residential properties, and all other securities, loans, and leases across maturities and repricing breakdowns listed in bank Call Reports as of Jan-2022. Data on bank asset holdings are available quarterly from Compustat's bank regulatory data. We then impute changes in the mark-to-market implied value of these securities and loans by matching them to the corresponding Treasury Indexes by maturity m. Treasury Index returns are from CRSP's "US Treasury and Inflation Indexes" data. As shown in Appendix Figure A.1, Treasury Bond Indexes with the longest maturity were most strongly affected by interest rate increases in 2022 and incurred the highest losses. Consequently, banks with a high share of long-duration assets incur the largest mark-to-market asset decline in our sample.

We estimate mark-to-market changes in the value of banks' assets between Q1-2022 and Q1-2023 as follows:

$$\begin{aligned}\Delta Loss\ MTM = \sum_m \Big(&(RMBS_m + Mortgages_m)\times \\ &\Delta Treasury\ Price_m \times Multiplier \Big) \\ &+ \sum_m \Big(Treasuries,\ securities,\ loans_m \times \Delta Treasury\ Price_m \Big),\end{aligned} \tag{1}$$

where m indexes the maturity and repricing breakdowns (i.e., 1 month, 3 months, 12 months, 2 years, 5 years, 7 years, 10 years, 20 years, and 30 years). $\Delta Treasury\ Price_m$ is the market price change (in %) of Treasury Bonds corresponding to maturity m from Q1-2022 to Q1-2023. To account for repayment risk in RMBS and mortgages, we construct the real-estate $Multiplier$ as the 2022 calendar year return of the iShares MBS ETF (MBB) scaled by the return of the overall treasury bond market (S&P U.S. Treasury Bond Index, ^SPBDUSBT), analogous to Jiang et al. (2024a).

We impute a bank's mark-to-market (MTM) implied asset value in Q1-2023 as the Q1-2022 value plus $\Delta Assets\ MTM$ from Eq. (1), and define the variable "% Asset Decline MTM" as the % change in asset value from Q1-2022 to Q1-2023-MTM.[10]

[9] Drechsler et al. (2024) develop refined measures of the uninsured deposit franchise based on uninsured share and uninsured deposit beta.

[10] To alleviate the concern that mismeasurement of *% Loss MTM* drives the patterns in our main findings, Appendix Table A.6 reports estimates of the main specification, dropping banks where the market-based measure may have more bias, i.e., the highest RMBS exposure and Commercial Real Estate (CRE) exposure. For example, RMBS have significantly shorter durations than Treasuries, conditional on refinancing maturity, which could lead to larger mismeasurement for banks that have more RMBS. In addition, we hand-collect changes in the fair values of bank assets as reported by the banks themselves. For this robustness test, we search for the 'carrying value' (CV) and 'fair value' (FV) of total financial assets in banks' 2022 10K-filings, and construct the implied asset value loss as $(FV - CV)/CV$. The results, summarized in Table



Appendix Figure A.2 shows the distribution of $\Delta Loss\ MTM$ across all FDIC-insured banks (Fig. A.2a) and across publicly listed bank holding companies (BHCs) (Fig. A.2b), which are aggregated at the bank holding company (BHC) level for BHCs with multiple FDIC-insured depository institutions. The range and mean of the distribution are comparable to the calculations in Jiang et al. (2024a) for both the full sample of banks and the publicly listed BHCs. We find a mark-to-market loss in asset value of about 17% for Silicon Valley Bank (SIVB) which is in the bottom quartile but not an extreme outlier. Appendix Figure A.4 displays the distribution of '% Loss MTM' across bank size (i.e., asset value) percentiles. We find that medium-sized institutions (60th to 90th percentile) exhibit the highest 'balance sheet risk' due to asset value declines.[11]

2.3. Stock price data

The stock price data are from FirstRate Data, a provider of intraday stock trade data. From this data set, we obtain intraday stock price and volume data for bank stock tickers in intervals of one, five and thirty minutes. For each time interval, the data set includes the price of the first and last trade, the highest and lowest prices, and the trade volume. Prices are adjusted for both splits and dividends. We use bank stock price data to compute the cumulative change in stock prices from March 1st to March 14th as well as hourly stock returns.

We also use these data to examine high frequency price changes around precise tweet timestamps during trading hours. More specifically, we adapt the methodology of Bianchi, Gomez-Cram, and Kung (2024). To do this, we use the timestamp (in Eastern Time) of every tweet t to determine two time intervals: one from 5 min before the tweet to *just before* the tweet and another from 5 to 15 min *after* the tweet. Then, for each bank, we identify the price of the last trade in the $[t-5, t)$ interval, denoted P_t, and the price of the first trade in the $[t+5, t+15]$ interval, denoted P_{t+5}. Following Bianchi et al. (2024), we define log returns as the difference in the logs of those prices. If there is no trading volume in both of these intervals, we exclude the tweet from the sample.

The timeline of the analysis is depicted below:

where $p_{i,t} = \log(P_{i,t})$ is the logged stock price, $p_{i,t}$ is the last logged stock price observed in the [−5,0] minute window, $p_{i,t+5}$ is the first logged stock price observed in the [5,15] minute window, and $\Delta p_{i,t} = \log(P_{i,t+5}) - \log(P_{i,t})$ is the log return.

The advantage of this high-frequency approach is that it is unlikely that other value-relevant information about the stock price becomes public during the short interval around the tweet. By estimating the log return using a starting price that was observed just *before* the tweet was posted and an ending price after, we reduce the likelihood that our findings are driven by reverse causality, i.e., that Twitter activity is reacting to stock price changes. However, to mitigate concerns that the tweets and prices jointly react to recent price movements and recent news at high frequency, our high frequency specifications control for recent price changes $\Delta p_{(-10,-\epsilon)} = \log(P_{i,t}) - \log(P_{i,t-10})$, as well as the number of articles posted (and their RavenPack ESS sentiment) in the 10 min prior to the tweet.

2.4. Empirical approach and twitter pre-exposure

Our main sample spans from January 1, 2023 through March 14, 2023. We have comprehensive Twitter data and have high-frequency stock return data from FirstRate through March 14. Our full Twitter sample covers 359 banks with non-zero Twitter attention in the pre period, but not all of these banks have market and balance sheet data available. The merge with run period returns data from FirstRate drops some sparsely traded bank stocks, leaving 335 observations. Merging with our key balance sheet variables (*% Loss MTM* and *% Uninsured*) requires information on security holdings, further reducing the sample size. Our final cross-sectional sample contains full information on 277 banks and bank holding companies. Appendix A.I presents these sample composition details, plus additional information on the underlying linking procedure between our data sources.

Fig. 3 portrays the timeline of our sample period and the timing of our empirical strategy. Our cross-sectional tests estimate how exposure to Twitter relates to run severity, proxied by bank stock losses in the run period. To measure *Twitter pre-exposure*, we compute the logged number of tweets about each bank's cashtag (e.g., $SIVB) in a pre-run period from January 1, 2023 through February 15, 2023. Throughout our analysis, this main variable is winsorized at the 95th percentile to limit the influence of outliers in our tests. We obtain similar findings throughout the paper if we winsorize at the 99th percentile or not at all. We also consider robustness to alternative pre-periods for our main tests (see Table A.4, Panels b and c). The distribution of Twitter pre-exposure has a well-behaved, single-peaked distribution, illustrated in Fig. 4.

Ideally, our empirical strategy would rely on exogenously assigned Twitter attention to bank stocks just prior to the run; however, given the networked nature of social media and the non-random choices of who participates in it, exogenous exposures to social media are elusive. Given this empirical challenge, we focus on pre-period Twitter attention as the key variable of interest, which has useful variation in the context of the SVB run. More concretely, our rationale for using pre-period Twitter attention is two-fold. First, the content of Twitter in the pre-run period is unrelated to regional banking risks: The most common words in the pre-period have nothing to do with banking stability, unlike the conversation that emerged during the week of the SVB run (see Figure A.5 in the Appendix).[12] Second, attention on FinTwit is persistent over time. This attention typically reflects investor fanaticism (Pedersen, 2022), which generates Twitter exposure that is unrelated to bank run risks. Because it is central to the credibility of our cross-sectional tests, we provide greater detail on each of these points below.

While Pedersen (2022) introduced the terminology of "fanatics" in a theoretical context, empirical evidence in the financial social media literature more broadly supports the idea that fanaticism is pervasive in

A.7, confirm our main findings. The two measures are positively correlated. The scatterplot of MTM-implied and FV asset losses in Appendix Figure A.3 illustrates the significant positive correlation ($r = 0.36$) between the two measures.

[11] A comparison of the banks with social media coverage in our sample to the remaining publicly traded bank holding companies (BHCs) (Appendix Table A.3a) and to the universe of call-report filing institutions (Appendix Table A.3b) reveals that (a) our sample includes over 80% of publicly traded BHCs, (b) is similar in terms of size (total assets, equity, total deposits), growth opportunities (M/B), and % Uninsured to public BHCs without social media coverage, and (c) is similar in terms of 2022% MTM Losses but much larger than non-public banks, on average.

[12] For the contextual analysis in Figure A.5, we compute the word frequency distribution for this 46-day pre-run period, and contrast it with the 5 day run-period (March 8 through March 13).

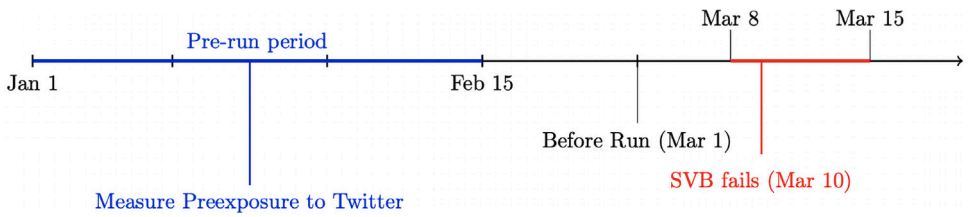


Fig. 3. *Timeline of SVB Run and Illustration of Empirical Strategy.* This figure presents a timeline that spans our main sample period from January 1 through March 15. It also illustrates definitions we use throughout the paper for pre-run period (January 1 – February 15) and run period (March 8 – March 15). Our main cross-sectional empirical strategy measures pre-exposure to Twitter by the number of original tweets authored about the bank's in the pre-run period and relates this pre-exposure to bank stock declines from March 1 (before the run) to the end of the run period. Figures A.5 and A.6 validate that tweet volume in the pre-period contains scant forward looking information in either its content or retweet patterns. Table A.4 Panels b and c, show robustness to alternative pre-periods.

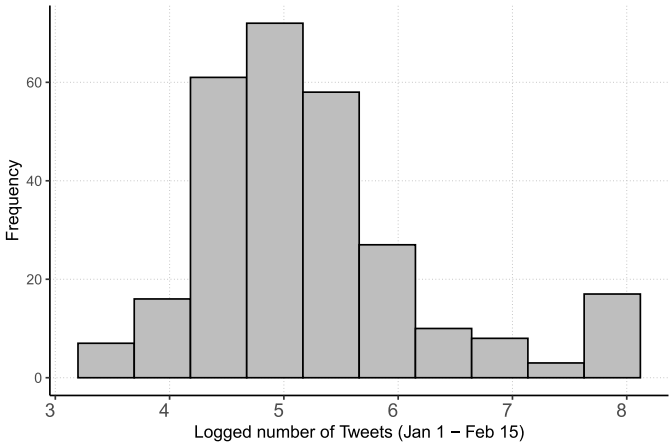


Fig. 4. *Distribution of Twitter Pre-Exposure.* This figure presents a histogram, depicting the cross-sectional distribution of Twitter pre-exposure. Twitter pre-exposure is the natural log of the number of cashtagged tweets about bank *i* posted between January 1, 2023 and February 15, 2023. The distribution is winsorized at the 95th percentile.

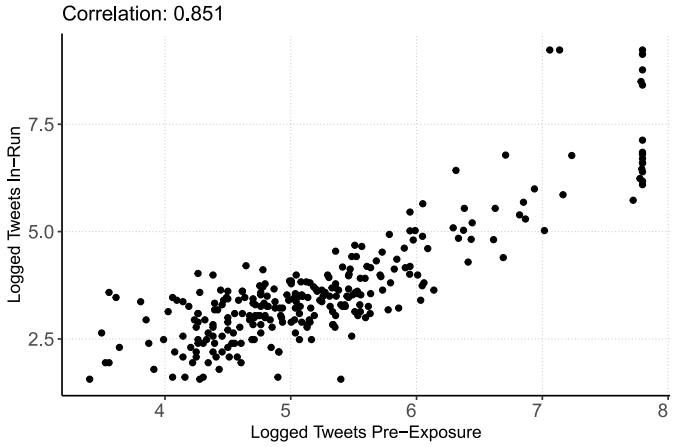


Fig. 5. *Twitter Exposure Pre- vs In-Run Period.* This figure presents a scatterplot that relates the logged number of Tweets in the pre-run period (January 1 – February 15) to the logged number of tweets in the run period (March 6 – March 14). As in our main Twitter pre-exposure measure, both distributions are winsorized at the 95th percentile. The correlation between these two variables is 0.85.

investor social media (Cookson et al., 2024b). For example, fanaticism implies that users tend to give more attention to stocks they like, to focus attention on the same stocks over time (Cookson et al., 2023), and to use an overall bullish tone (e.g., see Cookson and Niessner, 2020; Dim, 2025; Cassella et al., 2023; Müller et al., 2023). For this reason, we expect that bank stocks that are widely discussed on Twitter in January are still likely to be commonly discussed on Twitter in March. This initial discussion *among investors* creates a focal point for conversations during the run period. We find that our data are consistent with this idea: Twitter pre-exposure is both widely varying and strongly related to the number of tweets during the run (Fig. 5 illustrates this strong correlation $r = 0.85$).[13]

Although Twitter attention is not randomly assigned to bank stocks, cross-sectional variation in Twitter pre-exposure is unrelated to fundamental bank run risks. It would be problematic if pre-run Twitter attention reflected investor skepticism about bank stocks or strongly correlated with bank run risks. However, as Panel (a) of Fig. 6 illustrates, there is a weak *positive* correlation between sentiment about bank stocks and attention to bank stocks in the pre-run period ($r = 0.18$).[14] For the bank stocks in our sample, this positive correlation is consistent with fanaticism driving attention rather than skepticism. In addition, in Panel (b) of Fig. 6, we observe a near-zero correlation between a bank's run risk (i.e., *% Uninsured* × *% Loss MTM*) and logged number of tweets during the run period ($r = 0.023$). That is, variation in Twitter pre-exposure is empirically unrelated to fundamental bank run risks.

In Panel A of Table 2, we present the cross-bank distribution of total number of tweets in the pre-run period and run period, rescaled to be the number of tweets about a bank per 30 days. On average, the run period has more than four times as many tweets per 30 days than the pre-run period (2279 versus 537). There is substantial variation in Twitter pre-exposure: For example, the 10th percentile bank has only 33 tweets per 30 days while the 90th percentile bank has 511 tweets per 30 days written about it during the pre-run period. This distribution is right skewed, with a mean of 536 tweets per 30 days and a median

[13] This high correlation is not due to information from the pre-run period being repeated during the run period. Even for tweets that are retweeted at least once, less than 5% of original tweets from the pre-run period are *ever* retweeted during the run period (see Appendix Figure A.6a). A tweet thread by an account called 'Raging Capital Ventures' is the most notable exception (identified by listing the tweets with the most retweets in our sample, see Table A.2), but even this tweet thread was shared mostly after the run began—i.e., it was a reflection of information sharing during the run, and unlikely a cause of the run itself (see Figure A.7b). During the run, sharing of misinformation did happen (e.g., see Figure A.8b), but these examples are difficult to identify, even ex post, given the self-fulfilling nature of bank runs. Further, the tight relation between Twitter pre-exposure and in-run tweet activity remains even after controlling for our full suite of control variables, as shown in Appendix Figure A.9.

[14] This correlation is somewhat smaller after controlling for other bank characteristics (see Appendix Figure A.10).

Table 2
Summary statistics on tweet activity.

Panel A: Distribution of tweets per 30 days across banks – Run and Pre-Run Periods									
	Mean	Std	Min	10%	33%	50%	66%	90%	Max
Pre-run period total tweets	536.74	2207.19	1.96	33.26	60.65	88.70	136.30	511.30	23 478.91
Run period	2278.98	23 178.72	6.00	12.00	42.00	66.00	120.00	732.00	415 746.00

Panel B: Top Five Banks by "Run Behavior" Dictionary			
	Run	Contagion	Tweets Pre-Run
SIVB	6528	9662	1163
FRC	1249	1368	1257
SI	343	342	20,774
SBNY	260	106	2403
JPM	206	245	30,063
90th percentile	3	2	784

Panel C: Cross-Correlations of Twitter Media Pre-Exposure and In-Run Tweet Activity						
	Exposure Pre-Run	Contagion In-Run	Run In-Run	Startup In-Run	Tweets In-Run	Run Risk (z)
Exposure Pre-Run	1	.	.	.	.	.
Contagion In-Run	.578	1	.	.	.	.
Run In-Run	.650	.921	1	.	.	.
Startup In-Run	.795	.784	.830	1	.	.
Tweets In-Run	.850	.772	.815	.955	1	.
Run Risk (z)	.023	.208	.228	.201	.168	1

This table presents summary statistics on the number of tweets at the bank level. Panel A presents the number of tweets about a bank's cashtag for the pre-run period (Jan 1 through Feb 15, 46 days) and the run period (Mar 8 through Mar 13, 5 days) on a per-30-days basis. Panel B presents the counts of tweets by bank for "Run" behavior during the run period, "Contagion" tweets during the run period, and the total number of tweets in the pre-run period ("Tweets Pre-Run"). This panel depicts the top 5 banks by "run" tweet count during the run period. For comparison, we also compute the 90th percentile across banks. Panel C shows the cross-correlations of Twitter pre-exposure (i.e., the number of tweets in the pre-run period), in-run tweets that mention contagion, run, and startup words, and total in-run tweets, respectively, as well as 'run risk (z)', i.e., the product of *% Uninsured* and *% Loss MTM*, scaled to have mean zero and standard deviation of one. Throughout this table, the pre-run period is Jan 1 through Feb 15, 2023. The run period is defined as Mar 8 through Mar 13.

of only 88.7 tweets per 30 days. Taking the log of the number of tweets in the pre-period and winsorizing at the 95th percentile leads our main measure of Twitter pre-exposure to have a well behaved distribution (Fig. 4).

Next, we validate the contextual dictionaries in Table 1 by identifying the top 5 banks, sorted by counting "run" tweets posted in the run period. Panel B in Table 2 presents counts of run tweets, contagion tweets and pre-run tweets for these top 5 banks, as well as the 90th percentile of these counts in our bank sample. This ranking identifies SIVB as the bank ticker with the most run-based conversation, with 6528 "run" tweets. First Republic Bank (FRC), which also faced notable troubles during the run period (and subsequently failed on May 1, 2023), was next in this list with 1249 "run" tweets. All of these high run banks have an abnormally high number of contagion tweets, and all five banks are well above the 90th percentile for number of pre-run tweets (784).

These summary statistics highlight that the banks that faced the greatest run period distress also tended to have the most Twitter pre-exposure. The lone exception to this pattern is $JPM (JP Morgan Chase). Unlike regional banks, JPM did not experience distress during this period (rather (Caglio et al., 2023) document that the largest banks experienced deposit inflows), but they have many tweets written about them at all times. To mitigate this concern and the more general concern that large banks are more resilient to distress (Baron et al., 2023), we control for bank size in our empirical tests, and we also consider tests that exclude 'too big to fail (TBTF)' banks, following (Caglio et al., 2023)'s definitions.

2.5. Determinants of twitter pre-exposure

Our initial evidence indicates that Twitter pre-exposure is determined in part by investor fanaticism and is broadly unrelated to traditional run risks. However, other bank characteristics, the bank's information environment and financial characteristics might influence attention to bank stocks in the pre-run period while also relating directly to run risks. To understand which characteristics might confound inferences about Twitter's role, a comprehensive understanding of how Twitter pre-exposure relates to these characteristics is crucial. We provide such an evaluation in this section.

Specifically, we estimate cross-sectional specifications of the following form:

$$\begin{aligned} Twitter\ Pre\text{-}Exposure_i = a + b_1 \times Size_i + b_2 \times \%LossMTM_i \\ + b_3 \times \%Uninsured_i \\ + b_4\mathbf{X}_i + b_5\mathbf{W}_i + b_6\mathbf{Z}_i + \epsilon_i, \end{aligned} \tag{2}$$

where Twitter $Pre\text{-}Exposure_i$ is the log number of tweets about bank i between January 1 and February 15, winsorized at the 95th percentile, $Size_i$ is the logged Market Capitalization of the bank stock (as of December 31, 2022), *% Loss MTM* is the percent mark-to-market losses for bank i, *% Uninsured* is the percent of bank i deposits that are uninsured according to the Q4-2022 Call Reports, $\mathbf{X}_i$ is a vector of controls for the information environment, including analyst coverage computed from I/B/E/S and Dow Jones Newswire Coverage from Ravenpack, $\mathbf{W}_i$ is a vector of market controls (Q3-2022 market returns, Q4-2022 market returns, the sum of 3-day cumulative abnormal returns around 2022 FOMC announcements, the Q4-2022 spike in attention to bank i's ticker via Google searches), and $\mathbf{Z}_i$ is a vector of other bank controls, capturing deposit flows in Q4-2022, geographic footprint (# of Zipcodes with branches), deposit beta as in Drechsler et al. (2021), and the bank's market-to-book ratio. Appendix Table A.1 provides details on the construction of each of these variables.

Table 3 presents the results from estimating Eq. (2). The variable with the strongest relation to Twitter pre-exposure is *Size* or $log(MCap)$, which is positive and highly statistically significant across all specifications. Appendix Figure A.11 presents a scatterplot that highlights both the strength and linearity of this relationship. By contrast to size, traditional run risk factors like *% Uninsured* and *% Loss MTM* bear

Table 3
Determinants of twitter pre-exposure.

	Social media pre-exposure					
	(1)	(2)	(3)	(4)	(5)	(6)
ln(MCap) (z)	0.767***	0.730***	0.738***	0.744***	0.691***	0.592***
	(0.043)	(0.047)	(0.054)	(0.050)	(0.074)	(0.077)
% Loss MTM (z)	−0.027	−0.025	−0.026	−0.066*	−0.085**	−0.070*
	(0.040)	(0.039)	(0.039)	(0.037)	(0.037)	(0.036)
% Uninsured (z)	−0.051	−0.056	−0.055	−0.074	−0.067	−0.030
	(0.050)	(0.051)	(0.051)	(0.046)	(0.046)	(0.048)
Log N Articles (z)		0.101**	0.111*	0.087	0.045	0.037
		(0.043)	(0.058)	(0.053)	(0.054)	(0.052)
Log N Analysts (z)			−0.019	−0.054	0.004	0.015
			(0.071)	(0.072)	(0.075)	(0.070)
Ret 2022-Q4 (z)				−0.010	−0.009	0.009
				(0.042)	(0.041)	(0.040)
Ret 2022-Q3 (z)				−0.078*	−0.075**	−0.064*
				(0.040)	(0.038)	(0.038)
FOMC 2022 CAR (z)				−0.183***	−0.170***	−0.125***
				(0.050)	(0.045)	(0.047)
Δ SVI Q4-Q3 (z)				0.052	0.043	0.047
				(0.047)	(0.049)	(0.046)
% Dep Flow 2022-Q4 (z)					−0.042	0.103
					(0.053)	(0.085)
N ZIPs Branches (z)					0.064	0.063
					(0.044)	(0.043)
Deposit Beta (z)					−0.025	−0.035
					(0.038)	(0.037)
MTB BHC (z)					−0.083**	−0.036
					(0.034)	(0.034)
% CRE Loans (z)						−0.141***
						(0.041)
% Liquid Assets (z)						0.061
						(0.058)
Deposit Concentration (z)						0.071***
						(0.019)
Constant	5.235***	5.235***	5.235***	5.234***	5.234***	5.223***
	(0.038)	(0.038)	(0.038)	(0.036)	(0.035)	(0.035)
Observations	277	277	277	277	277	274
R^2	0.588	0.597	0.597	0.643	0.654	0.677
Adjusted R^2	0.583	0.591	0.590	0.631	0.637	0.657

This table presents OLS regressions that relate Twitter Pre-Exposure to characteristics of the bank, market and information environment that could drive attention to banks. The dependent variable is 'Twitter pre-exposure', i.e., the log of the number of tweets about a given bank in the pre-run period from Jan 1 through Feb 15, 2023. The explanatory variables include bank characteristics such as size (ln(MCap)) and valuation (MTB), run risk variables (% Loss MTM and % Uninsured depositors), attention characteristics (analyst coverage and number of newspaper articles in 2022, change in Google Search Volume SVI between Q4 and Q3 of 2022), stock returns (in Q3 and Q4 of 2022 and around 2022 FOMC meetings), measures of the deposit franchise (i.e., deposit beta from Drechsler et al. (2017) and the number of unique states with bank branches), and measures of deposit characteristics (% of commercial-real-estate (CRE) loans, deposit flow in Q4-2022, supply-driven deposit inflows in 2022, and deposit concentration). All variables denoted with '(z)' are standardized to have a mean of zero and standard deviation of one to ease magnitude comparisons.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

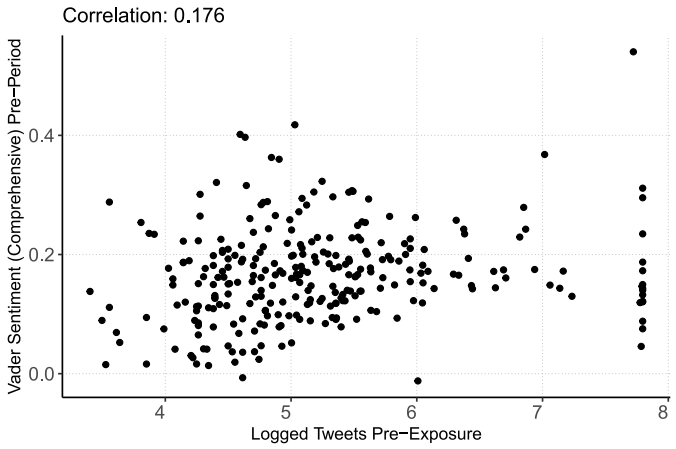
*** Indicate statistical significance at the 1% level.

a weak relation to Twitter pre-exposure. If anything, higher losses and uninsured deposits are slightly *negatively* related to Twitter pre-exposure in some specifications, which contrasts with the view that Twitter pre-exposure proxies for run risk. This low correlation between aspects of run risk and Twitter pre-exposure provides support for our empirical strategy, which predicts run severity using interactions between Twitter pre-exposure and run risk factors.

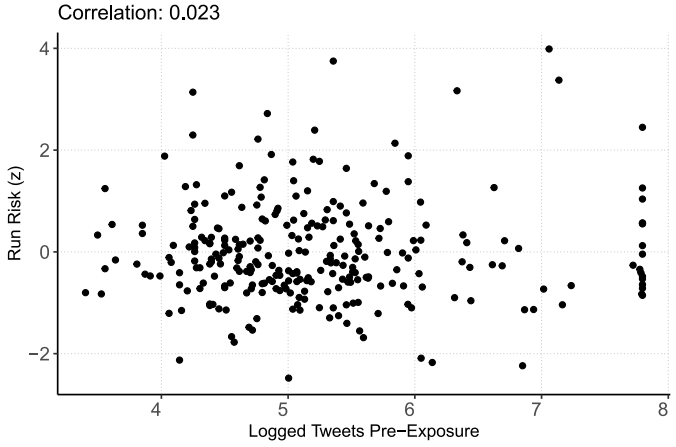
Other characteristics relate to Twitter pre-exposure, but not nearly as strongly as size. For example, the number of traditional news articles is positively related to Twitter pre-exposure, significant across most specifications. Additionally, banks with low stock market returns tend to have more Twitter pre-exposure (see the coefficients on Ret Q3-2022, FOMC 2022 CAR, and in column 5, market-to-book ratio).[15] However, other characteristics are unrelated to Twitter pre-exposure. For example, Q4-2022 spikes in retail investor attention (Δ SVI), deposit beta, geographic footprint, and number of analysts covering the bank stock are all statistically insignificant.

Because some of the characteristics that relate to Twitter pre-exposure could relate to run risks directly, particularly bank size, throughout the paper, we consider specifications that control for these characteristics. This approach is informative of Twitter's role as long as there is no unobserved bank characteristic that predicts run risk and relates to Twitter exposure that is not captured by the variation in our set of control variables.

[15] In column 6, CRE Loans (%) is negatively related to pre-exposure and deposit concentration is positively related to pre-exposure, both statistically significant. Unlike the factors described in the main text, there is not an obvious rationale for these factors driving attention to bank stocks. For example, CRE Loans were a significant a driver of bank losses before the run (Jiang et al., 2025), but they are *negatively* related to pre-exposure.



(a) Twitter Pre-Exposure vs Vader Sentiment in Pre-Period



(b) Twitter Pre-Exposure vs Run Risk.

Fig. 6. *Twitter Pre-Exposure vs Tweet Sentiment and Run Risk*. This figure presents scatterplots depicting the relation between Twitter pre-exposure (x-axis) and either VADER sentiment of tweets in the pre-period (Panel (a)) or 'run risk' (Panel (b)) where run risk is *% Uninsured* × *% Loss MTM*, measured in Q4 2022. The correlation in Panel (a) is 0.18 and it is 0.02 in Panel (b).

3. Results

In this section, we examine the link between social media and bank run severity during the SVB run period. We conduct two types of tests. First, we relate Twitter *pre-exposure* to bank run severity, measured using run period bank stock losses for publicly traded banks and Q1-2023 deposit outflows. Second, *within the run period*, we relate social media signals (i.e., Twitter attention and sentiment) to returns at high frequency. The high frequency tests are conducted at the hourly level, and separately, at the tweet level, adapting the methodology of Bianchi et al. (2023).

Before presenting our main tests, we validate the use of returns to proxy for run severity by splitting the sample into banks with balance sheet risk above and below the median, where balance sheet risk is *% Loss MTM* × *% Uninsured*. Fig. 7 presents the cumulative returns during the run period, separately for banks with high versus low run risk. We see that returns sharply decline at the onset of the run on March 9 without much loss in the days before. Consistent with in-run returns reflecting run severity, the decline is greater for banks with high run risk.[16]

3.1. Twitter pre-exposure and run period losses

The cross-sectional tests in this section follow the empirical strategy motivated in Section 2.4. In these tests, we examine how Twitter *pre-exposure* interacts with *% Loss MTM* and *% Uninsured*. To measure Twitter pre-exposure, we compute the log of the number of 'cashtagged' tweets about each bank between Jan 1, 2023 and Feb 15, 2023, winsorized at the 95th percentile. As we show in Section 2.4, Twitter pre-exposure does not generally contain discussions of banking distress, and it is unrelated to traditional bank run risks (see Fig. 6). As a first approximation, therefore, we attribute estimated effects of Twitter pre-exposure to the bank's *exposure to social media* rather than information contained in social media about how run prone one bank is versus another. To strengthen this inference, we also estimate specifications that control for market, bank, and funding characteristics included in the determinants regressions reported in Table 3.

[16] There is information in each component of run risk (*% Losses MTM* and *% Uninsured*). See Appendix Table A.9 for four-way interaction tests that illustrate this. However, for simplicity of exposition, we primarily report results based on the combined run risk variable. As an external point of validation, Cipriani et al. (2024) finds that the vast majority of banks that experienced a run around SVB's failure (18 out of 22) were publicly traded, suggesting an important role for public pressure through equity market attention.

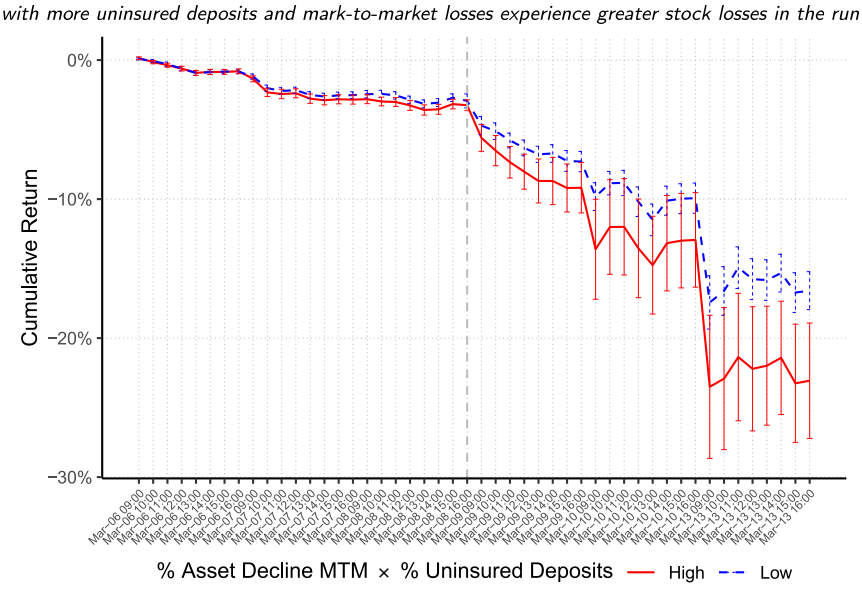


Fig. 7. *Cumulative Returns of Banks with High versus Low Run Risk.* This figure shows the average cumulative returns of US publicly listed banks and bank holding companies (BHCs) from March 06 through March 13, 2023. The sample is split by "Run Risk" defined as the % of uninsured deposits × the % MTM-implied asset value decline. 'High' represents the top half above the median (red solid line), 'Low' represents the bottom half (blue dashed line). The gray vertical dashed line indicates March 09, 2023 at 9am. 90% confidence intervals around the mean cumulative return are constructed by bootstrapping with clustering at the bank level.

Specifically, we estimate a cross-sectional specification that interacts Twitter pre-exposure with run-relevant balance sheet characteristics for bank i:

$$Stock\ Loss_i = \beta_1 \%LossMTM_i \times \%Uninsured_i \times Pre\text{-}Exposure_i + lower\ order\ terms + \gamma \mathbf{X}_i + \epsilon_i, \quad (3)$$

where $Stock\ Loss_i$ is the percentage of bank i's stock market value that was lost from March 1st until March 14th. $Pre\text{-}Exposure_i$ is the log of the number of tweets in the pre-period, winsorized at the 95th percentile. The main coefficients of interest are those on terms involving $Pre\text{-}Exposure_i$. The coefficients on the interactions of run risk variables ($\%\ LossMTM_i$ and $\%\ Uninsured_i$) with $Pre\text{-}Exposure_i$ speak to the hypothesis that exposure to social media amplifies bank run risks. The coefficients on the lower order terms in the specification are also of interest as a benchmark for magnitudes. Finally, the vector $\mathbf{X}_i$ includes controls for drivers of retail attention, bank size, news coverage, 2022 bank stock returns, and bank funding characteristics. Appendix Table A.1 gives full detail on how these variables are constructed. Variables indicated with a '(z)' in the regression tables have been standardized to have a mean of 0 and standard deviation of 1 for comparability.

Table 4 presents the main results. Column 1 of Panel A of Table 4 presents a specification without Twitter pre-exposure. Interpreting the main effect, a standard deviation increase in *% Uninsured* is associated with a 4.249 percentage point decline in the bank's stock during the SVB bank run. In addition, consistent with a model of solvency runs as in Drechsler et al. (2024) and Jiang et al. (2024a), the interaction between *% Uninsured* and *% Loss MTM* is positive and significant. A standard deviation increase in *% Loss MTM* is associated with 1.742 percentage points more stock market losses during the run for a bank with high uninsured deposits (one standard deviation above the mean) versus a bank at the average of uninsured deposits.

In column 2 of Panel A of Table 4, we relate Twitter pre-exposure to bank stock losses during the SVB run in a simple regression without controls. A standard deviation increase in Twitter pre-exposure is associated with 4.317 percentage points greater stock market loss during the SVB run, a magnitude similar to a standard deviation increase in *% Uninsured*. When we control for traditional bank run risks in column 3, we estimate similar magnitudes on Twitter pre-exposure.

In coordination models of bank runs, exposure to social media could affect bank run severity through amplifying existing risks (Goldstein and Pauzner, 2005). While a no-run equilibrium can be sustained with uninsured depositors, this is less likely when depositors communicate openly about running (Kelly and Gráda, 2000), making run risk amplification via social media a natural possibility. To evaluate this idea, we estimate a specification that interacts *Pre-Exposure*, *% Uninsured* and *% Loss MTM* in column 4. Consistent with amplification, we estimate significant positive interactions for $Uninsured \times Pre\text{-}Exposure$ and for $Loss \times Uninsured \times Pre\text{-}Exposure$. These estimates imply that banks with higher *% Uninsured* experience greater run severity *if* they have high Twitter pre-exposure. Moreover, the triple interaction implies that a bank with high run risk (one sd above the mean for *both* $\%LossMTM$ *and* $\%Uninsured$) can expect 2.407 percentage points more run-period stock market losses for a one standard deviation increase in Twitter pre-exposure.

Panel B of Table 4 repeats these main tests in specifications that include a comprehensive set of control variables. Upon including controls, the magnitudes dampen somewhat, but the main conclusions are robust. In the non-interactive specification in column 1, we note that the *% Loss MTM* variable becomes statistically insignificant while the double interaction $Uninsured \times Loss\ MTM$ remains statistically significant. In column 2, we similarly observe a smaller magnitude on Twitter pre-exposure than the unconditional estimate (1.922 versus 4.317), but this coefficient estimate remains highly statistically significant. Beyond the coefficients of interest, the controls have the expected sign and significance. For example, higher analyst coverage predicts more run losses. Also, returns from late 2022 and around 2022 FOMC announcements strongly and negatively predict run period losses, consistent with work by Choi et al. (2023). In addition, banks with greater geographic diversification, higher market-to-book ratio, and fewer deposit outflows in Q4-2022 tend to have lower run period losses.

Turning to the interactive specifications reported in columns 3 and 4, the triple interaction $Loss \times Uninsured \times Pre\text{-}Exposure$ remains statistically significant with a smaller magnitude compared to the

Table 4
Twitter pre-exposure and bank stock losses.

Panel A: Baseline result				
	% Stock value loss during run			
	(1)	(2)	(3)	(4)
% Uninsured (z)	4.249***		3.276***	2.715***
	(1.026)		(0.840)	(0.524)
% Loss MTM (z)	1.192		1.579*	0.227
	(0.879)		(0.814)	(0.499)
% Uninsured (z) × % Loss MTM (z)	1.742**		1.838**	0.778**
	(0.824)		(0.737)	(0.375)
Pre-Exposure (z)		4.317***	3.683***	3.724***
		(0.936)	(0.807)	(0.730)
Pre-Exposure (z) × % Uninsured (z)				2.695***
				(0.701)
Pre-Exposure (z) × % Loss MTM (z)				0.329
				(0.826)
Pre-Exposure (z) × % Uninsured (z) × % Loss MTM (z)				2.407***
				(0.561)
Constant	16.555***	16.112***	16.585***	15.666***
	(0.648)	(0.547)	(0.595)	(0.457)
Observations	277	277	277	277
R^2	0.190	0.185	0.312	0.430
Adjusted R^2	0.181	0.182	0.302	0.415

(*continued on next page*)

specifications without controls in Panel A of Table 4. Given the strong correlation between bank size and Twitter pre-exposure, we additionally include the interaction of $ln(MCap)$ with *% Uninsured* and *% Loss MTM*, as well as the lower level interactions, in these specifications. Although this triple interaction between bank size and run risk factors is significant, indicating greater salience of risk among larger financial institutions, the triple interaction of $\%LossMTM \times \%Uninsured \times Pre\text{-}Exposure$ continues to exhibit its own strong relation to run period losses. These findings imply that Twitter pre-exposure does not merely reflect other factors that significantly predict run severity.[17]

We perform several robustness exercises beyond those reported in the main text. First, we include interactions of each of the control variables with *% Uninsured* and *% Loss MTM*, and the corresponding lower-level interaction terms. We include these interactions one at a time to conserve on degrees of freedom given our relatively small cross-section of publicly traded banks, while continuing to include the full suite of controls (without interactions) in each estimation. Across all specifications with interacted controls, we continue to estimate a significant triple interaction between $Pre\text{-}Exposure$, $Uninsured$ and $Loss\ MTM$, supporting the interpretation that exposure to social media amplifies bank run risks. Appendix Figure A.12 presents point estimates and confidence intervals for these tests.[18] Second, one concern is that the largest banks (known to experience deposit *inflows*, see Caglio et al., 2023) are also the bellwethers of the banking industry, and thus, their stock prices may be more responsive to banking industry distress unrelated to run reasons. To alleviate this concern, we re-estimate the main specification after dropping banks with more than $500 billion in deposits. The results are virtually identical to our main specification (see Appendix Table A.4a). Third, another natural concern is that the pre-run period from Jan 1 through Feb 15 is a period when tweet volume reflected banks with poor fundamentals that were likely to perform poorly during the run period — i.e., the pre-run Twitter exposure variable might not reflect pure social media exposure, but also information about bank fundamentals. The evidence in Section 2.4 addresses this concern directly, as does the inclusion of controls in Panel B of Table 4. However, we further alleviate this concern by re-estimating the main specification with two alternative pre-exposure periods: November 15 – December 31, and October 1 – November 15. The results closely mirror the estimates based on our main pre-period (see Appendix Tables A.4b and A.4c).[19]

3.1.1. Nature of in-run tweets

Next, we investigate how Twitter pre-exposure could lead to heightened attention and discussion of bank run topics during the run period. For this analysis, we estimate a cross-sectional test with the run period bank stock market loss as the dependent variable, and then we enrich the specification in Eq. (3) by controlling for counts of run-related tweets *during* the run period. Specifically, we use the 'run behavior' and 'contagion' content dictionaries to identify run-relevant tweets during the run period. We also account for the number of 'tech community' tweets during the run period.

Panel C of Table 4 presents the results from estimating these specifications. For comparison, column 1 repeats our main specification by reporting the regression without any controls for run-period tweet counts. Upon controlling individually for contagion tweets, run tweets, and tech community tweets during the run period, individually (columns

[17] We also estimate a version of the main specification with indicators of the top terciles of *% Uninsured*, *% MTM Loss*, and *Pre-Exposure* instead of the continuous measures (see Appendix Table A.4d), allowing these variables to enter the specification non-linearly. These tests yield similar insights, alleviating concerns that the *% Uninsured* × *Pre-Exposure* interaction proxies for a nonlinear effect of uninsured deposits.

[18] Beyond this, we also estimate a specification that includes all of these interactions together in the same specification. The triple interaction $Loss\ MTM \times Uninsured \times Pre\text{-}Exposure$ remains statistically significant in this specification, as shown in column 4 in Appendix Table A.5. We also perform a specification curve analysis following (Simonsohn et al., 2020) in Appendix Figure A.13, in which we estimate *all* possible combinations of these interactive specifications (16,383 in total, i.e., 2^{14} minus the empty specification). This analysis reveals a high degree of robustness: Compared to a benchmark in which the *Pre-Exposure* variable is permuted, roughly 90% of these specifications deliver an estimate above the 95th percentile of the distribution of permuted estimates. Moreover, nearly all of the triple interaction coefficients of interest are positive.

[19] We additionally also estimate our main specification in Eq. (3) using changes in fair asset values as disclosed in banks' 10K-filings (see Section 2.2 for the construction of the data) instead of the market-implied measure of Asset Losses. The results remain qualitatively similar as shown in Appendix Table A.7.

Table 4 (*continued*).

Panel B: Including controls for size, valuation, attention, prior stock returns, deposit flows, deposit franchise, risk				
	% Stock value loss during run			
	(1)	(2)	(3)	(4)
Pre-Exposure (z) × % Uninsured (z) × % Loss MTM (z)			1.437***	1.660***
			(0.354)	(0.424)
Pre-Exposure (z) × % Uninsured (z)			1.862***	1.641***
			(0.530)	(0.613)
Pre-Exposure (z) × % Loss MTM (z)			−1.080*	−0.276
			(0.563)	(0.632)
Pre-Exposure (z)		1.922**	2.339***	1.649**
		(0.744)	(0.608)	(0.643)
% Uninsured (z) × % Loss MTM (z)	1.794**		0.541	0.710*
	(0.726)		(0.390)	(0.396)
% Uninsured (z) × % Loss MTM (z) × ln(MCap) (z)			1.627***	1.457***
			(0.400)	(0.404)
% Uninsured (z)	2.224***		2.306***	2.471***
	(0.664)		(0.471)	(0.477)
% Uninsured (z) × ln(MCap) (z)			1.354**	1.381**
			(0.586)	(0.584)
% Loss MTM (z)	0.693		−0.677	−0.383
	(0.690)		(0.464)	(0.538)
% Loss MTM (z) × ln(MCap) (z)			2.408***	1.770***
			(0.643)	(0.617)
ln(MCap) (z)	1.657	1.308	−0.565	−0.458
	(1.048)	(1.379)	(0.867)	(0.860)
Log N Analysts (z)	1.644**	1.921**	1.998***	2.262***
	(0.739)	(0.761)	(0.726)	(0.724)
Log N Articles (z)	−0.260	−0.512	−0.460	−0.649
	(0.595)	(0.616)	(0.550)	(0.545)
Ret 2022-Q4 (z)	−1.659***	−1.928***	−0.754*	−0.411
	(0.527)	(0.649)	(0.423)	(0.428)
Ret 2022-Q3 (z)	−0.709	−0.581	−0.167	−0.259
	(0.588)	(0.662)	(0.446)	(0.443)
FOMC 2022 CAR (z)	−2.185***	−1.780***	−2.152***	−1.674***
	(0.552)	(0.551)	(0.499)	(0.530)
Δ SVI Q4-Q3 (z)	−0.696	−0.671	−0.980**	−0.847**
	(0.515)	(0.531)	(0.406)	(0.409)
% Dep Flow 2022-Q4 (z)	−2.429***	−2.253***	−2.193***	−0.200
	(0.797)	(0.688)	(0.558)	(0.636)
N ZIPs Branches (z)	−1.048*	−1.583**	−1.227***	−1.295***
	(0.623)	(0.764)	(0.418)	(0.413)
Deposit Beta (z)	−0.086	0.215	0.584	0.389
	(0.557)	(0.560)	(0.403)	(0.402)
MTB BHC (z)	−1.323***	−1.276**	−1.211***	−0.971**
	(0.502)	(0.529)	(0.411)	(0.426)
% CRE Loans (z)				−0.557
				(0.449)
% Liquid Assets (z)				0.780
				(0.705)
Deposit Concentration (z)				0.813*
				(0.485)
Constant	16.560***	16.115***	15.422***	15.385***
	(0.550)	(0.464)	(0.416)	(0.412)
Observations	277	277	277	274
R^2	0.487	0.430	0.671	0.635
Adjusted R^2	0.460	0.404	0.644	0.599

(*continued on next page*)

2 through 4) or all together (column 5), the coefficient on Twitter pre-exposure diminishes in both economic and statistical significance. In column 6, when we include the full suite of funding, market, deposits, attention and valuation controls, we obtain similar findings. These results suggest that Twitter pre-exposure predicts bank run severity because it leads to more run and contagion tweets, as well as more in-run Twitter attention by tech community users. We revisit these mechanisms with more precise tests that relate lagged Twitter attention to hourly returns in the run period in Section 3.3.

3.2. Twitter pre-exposure and deposit flows

Next, we investigate how Twitter pre-exposure predicts Q1-2023 deposit outflows, measured as the change in uninsured deposits and total deposits reported in the FDIC Call Reports from the end of Q4-2022 to Q1-2023. Specifically, we estimate the following cross-sectional specification:

$$Deposit_Outflow_i = \beta_1 \%Loss\ MTM_i \times \%Uninsured_i \times Pre\text{-}Exposure_i + lower\ order\ terms + \gamma \mathbf{X}_i + \epsilon_i \quad (4)$$

where $Deposit_Outflow_i$ is the percentage loss of deposits for either uninsured deposits or total deposits — i.e., $Deposit_Outflow = 100 \times \frac{Deposits_{Q4\text{-}2022} - Deposits_{Q1\text{-}2023}}{Deposits_{Q4\text{-}2022}}$. The variables $\%Loss\ MTM_i$ and $\%Uninsured_i$ are computed using the data available as of Q4-2022. As in Eq. (3), $Pre\text{-}Exposure_i$ is the logged number of tweets containing bank i's cashtag during the pre-run period (Jan 1 through Feb 15), winsorized at the 95th percentile. The main coefficient of interest is β_1, which

Table 4 (*continued*).

Panel C: Controlling for tweet activity during run period						
	% Stock value loss during run					
	(1)	(2)	(3)	(4)	(5)	(6)
Pre-Exposure (z)	4.317***	0.431	−0.408	−1.376	−0.892	−0.129
	(0.936)	(0.516)	(0.589)	(1.110)	(0.617)	(0.618)
Contagion Tweets In-Run (z)		6.696***			3.104**	3.942***
		(1.333)			(1.362)	(1.140)
'Run' Tweets In-Run (z)			7.252***		3.594***	4.398***
			(1.374)		(1.302)	(1.203)
Tech-Comm. Tweets In-Run (z)				7.150***	1.339*	0.280
				(1.906)	(0.799)	(0.830)
Constant	16.112***	16.096***	16.121***	16.127***	16.112***	16.108***
	(0.547)	(0.434)	(0.435)	(0.483)	(0.426)	(0.343)
Observations	277	277	277	277	277	274
R^2	0.185	0.484	0.489	0.372	0.509	0.666
Adjusted R^2	0.182	0.481	0.485	0.368	0.502	0.643
Funding controls						✓
Market controls						✓
Deposits controls						✓
Attention controls						✓
Valuation controls						✓

This table presents OLS estimates of regressions that relate bank stock losses during the run period to Twitter pre-exposure, mark-to-market losses, and percentage of uninsured deposits (Panel A presents the baseline results). All variables denoted with '(z)' are standardized to have mean zero and standard deviation of one. Panel B is similar to Panel A but additionally controls for bank size and valuation, attention characteristics, stock returns, measures of the deposit franchise and measures of deposit characteristics. Variable definitions are as in Table 3, with full detail on their construction in Appendix Table A.1. In Panel C, total 'Run', 'Contagion' tweets during the run count the number of tweets with at least one word in the 'Run' and 'Contagion' dictionaries between March 8 and March 13. Tech community tweets are counted over the same period, but based on the number of tweets about the bank stock posted by users with at least one tech community term in their Twitter user description. Robust standard errors are reported in parentheses.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

captures how much greater the percentage decline in deposit outflows is for high-run-risk banks if they also have high Twitter pre-exposure. Because the dependent variable is $Outflow$, a positive coefficient reflects a more severe run. We include the same vector of controls $\mathbf{X}_i$ as in Panel B of Table 4. That is, as in the interactive specifications, we include $ln(MCap)$ and its interactions with run risk factors ($Uninsured_i$ and $Loss\ MTM_i$).

Table 5 presents the results from estimating Eq. (4). Consistent with tests using run period returns, the percentage of *uninsured* deposits is strongly related to deposit outflows. Unconditionally, a standard deviation increase in *% Uninsured* in Q4-2022 is associated with 3.31 percent more uninsured deposit outflows (col 1) and 1.37 percent more total deposit outflows (col 3). In addition, Twitter pre-exposure predicts an amplification of run risks revealed by Q1-2023 deposit outflows. The triple interaction term is positive and significant, indicating that banks with high bank run risk experience significantly greater uninsured deposit outflows in Q1-2023 if they have higher $Pre\text{-}Exposure$. The magnitude and significance strengthens upon including controls, implying that the estimates do not merely reflect other factors influencing outflows (e.g., we partly control for a deposits channel as in Drechsler et al. (2017) using lagged outflows of insured and uninsured deposits).

Consistent with the idea that uninsured deposits are especially run-prone (Diamond and Dybvig, 1983; Egan et al., 2017), the estimated coefficients are smaller and less robustly related to pre-exposure for *total* deposit outflows. Overall, this evidence from quarterly deposits is complementary to the bank stock return evidence in our main tests. Because these data are quarterly, we cannot precisely identify when exactly the deposit outflows occurred during Q1-2023. Thus, this evidence should be taken as suggestive that *run period* deposit outflows were sensitive to Twitter pre-exposure. Complementary with these tests, recent work by Caglio et al. (2023) indicates that most of the unusual deposit activity happened in the week of the SVB collapse using confidential regulatory data on weekly deposit flows from the Federal Reserve. These results corroborate our main interpretations that the run period stock market prices are informative of the run itself.

We present two additional tests relating to deposit outflows in Appendix Table A.8. First, in Panel A.8a, we ask whether the stock market losses during the run period, denoted '*% Run Loss*,' capture the run risks relevant to Q1-2023 deposit outflows. Consistent with this idea, we estimate that the interaction between '% Run Loss' and Twitter pre-exposure is significant and positively associated with uninsured deposit outflows in Q1-2023 (column 4). In addition, when we estimate a specification that considers a horse race between '*% Run Loss*' and balance sheet risks (uninsured deposits and mark-to-market losses), we find that run loss remains significant while the triple interaction between *Pre-Exposure*, *% Uninsured* and *% Loss MTM* becomes small and statistically insignificant. These findings help to validate our use of stock market losses to proxy for bank run severity by drawing a tighter connection to deposit outflows. Second, in Panel A.8b, we exclude too-big-to-fail banks, which tended to experience deposit inflows during this period (Caglio et al., 2023). The results remain similar to the main deposit outflow tests in Table 5. We continue to find similar results.

One limitation of these cross-sectional tests is that the tweets during the run period might not precede the return reactions or deposit outflows. To provide finer-grained insight into the timing of run period tweets and bank run severity, we next turn to examine the impact of tweet activity in a bank-hourly panel *during the SVB run period*. These tests focus exclusively on returns since hourly deposit flow data are unavailable.

3.3. Hourly bank stock returns and in-run twitter activity

In this section, we examine how hourly bank stock returns relate to lagged in-run Twitter attention (i.e., the number of tweets in the last four hours) conditional on run risk factors during the SVB run period. To do this, we split our sample into banks with above-median and below-media run risk, i.e., '% Uninsured' × '% MTM Losses'. For each

Table 5
Twitter pre-exposure and deposit flows.

	Deposit outflows Q1-2023 (%)			
	Uninsured		Total	
	(1)	(2)	(3)	(4)
% Uninsured (z) × % Loss MTM (z) × Pre-Exposure (z)	1.47**	1.78**	0.673	1.51***
	(0.746)	(0.890)	(0.536)	(0.476)
% Uninsured (z) × Pre-Exposure (z)	1.59	0.989	1.74**	0.897
	(0.993)	(1.03)	(0.697)	(0.547)
% Loss MTM (z) × Pre-Exposure (z)	1.79*	0.208	0.788	0.501
	(0.923)	(1.19)	(0.704)	(0.679)
Pre-Exposure (z)	0.973	2.03	0.974	0.512
	(1.12)	(1.29)	(0.968)	(0.839)
% Uninsured (z) × % Loss MTM (z)	−0.878	−1.24	0.112	−0.260
	(0.652)	(0.835)	(0.369)	(0.459)
% Uninsured (z)	3.31***	4.25***	1.37***	1.83***
	(1.17)	(1.42)	(0.502)	(0.542)
% Loss MTM (z)	0.609	0.081	0.829*	0.088
	(0.759)	(0.886)	(0.470)	(0.545)
ln(MCap) (z) × % Uninsured (z) × % Loss MTM (z)		0.683		0.201
		(0.896)		(0.580)
ln(MCap) (z) × % Uninsured (z)		2.23*		1.89**
		(1.35)		(0.728)
ln(MCap) (z) × % Loss MTM (z)		2.68*		0.804
		(1.55)		(0.778)
ln(MCap) (z)		−0.831		1.58
		(1.57)		(1.02)
Deposit Beta (z)		−0.010		−0.169
		(0.816)		(0.521)
N States Branches (z)		−2.72**		−1.83**
		(1.30)		(0.719)
FOMC 2022 CAR (z)		−1.08		−1.39**
		(0.806)		(0.565)
Ret 2022-Q4 (z)		1.30		0.655
		(0.969)		(0.584)
Ret 2022-Q3 (z)		−2.37**		−0.396
		(1.13)		(0.659)
Supply-driven Deposit Inflow 2020 (z)		1.46*		1.11**
		(0.805)		(0.452)
Dep Inflow Uninsured 2022-Q4 (z)		−1.30		−1.70*
		(1.76)		(1.01)
Dep Inflow Insured 2022-Q4 (z)		0.165		−1.30
		(1.66)		(1.07)
Constant	4.79***	4.46***	−0.915*	−1.29**
	(0.863)	(1.06)	(0.534)	(0.569)
Observations	275	262	275	262
R^2	0.088	0.247	0.086	0.399
Adjusted R^2	0.064	0.171	0.062	0.339
Attention controls		✓		✓
Other valuation controls		✓		✓

This table presents cross-sectional OLS regression estimates relating quarterly bank deposit flows to Twitter pre-exposure, *% Uninsured* and *% Loss MTM*. The dependent variable, 'Deposit outflows (%)' is the percentage change in bank deposits from the end of Q4-2022 to the end of Q1-2023, multiplied by −1 so that a higher number indicates larger outflows. Columns 1 and 2 use the uninsured deposits (above the FDIC threshold of $250K), columns 3 and 4 use all deposits to construct outflows. Columns 2 and 4 additionally include controls for bank valuation and outside attention, as well as controls for insured and uninsured deposit flows in Q4-2022, supply-driven deposit inflows in 2020, and measures of 2022 stock returns and the value of the deposit franchise, as in Table 4. Details on variable construction and data sources are provided in Appendix A.1. Robust standard errors are reported in parentheses.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

subsample, we then estimate the following specification at the hourly frequency:

$$r_{i,t} = a + b_1 \times 1(\geq \text{Mar } 09)_t + b_2 \times 1(\text{N Tweets High})_{i,t-1} + b_3 \times 1(\geq \text{Mar } 09)_t \times 1(\text{N Tweets High})_{i,t-1} + \eta \mathbf{X}_{i,t-1} + \delta_i + \gamma_t + \epsilon_{i,t}, \quad (5)$$

where $r_{i,t}$ is the hour t return for bank i measured in basis points, the indicator '$1(\geq \text{Mar } 09)_t$' equals one after March 9 at 9 a.m. (the onset of the run) and zero before then. Since the number of tweets at the hourly frequency is highly skewed with a large number of zeros, we use an indicator variable to capture Twitter activity in this specification: '$1(\text{N Tweets High})_{i,t-1}$' is a dummy variable that takes the value of one, if the number of cashtagged tweets about bank i over the past 4 *hours* is above the median (within the subsample), and zero otherwise.[20] We set missing returns observations to zero and restrict the sample to the week of March 6th to March 10th.

This specification contrasts hourly returns for banks with high versus low recent Twitter attention, and it does so separately for banks with high versus low run risk during the run period. All specifications include a vector of controls $\mathbf{X}_{i,t-1}$, to account for traditional news coverage of bank i and bank i's price movements during the lagged

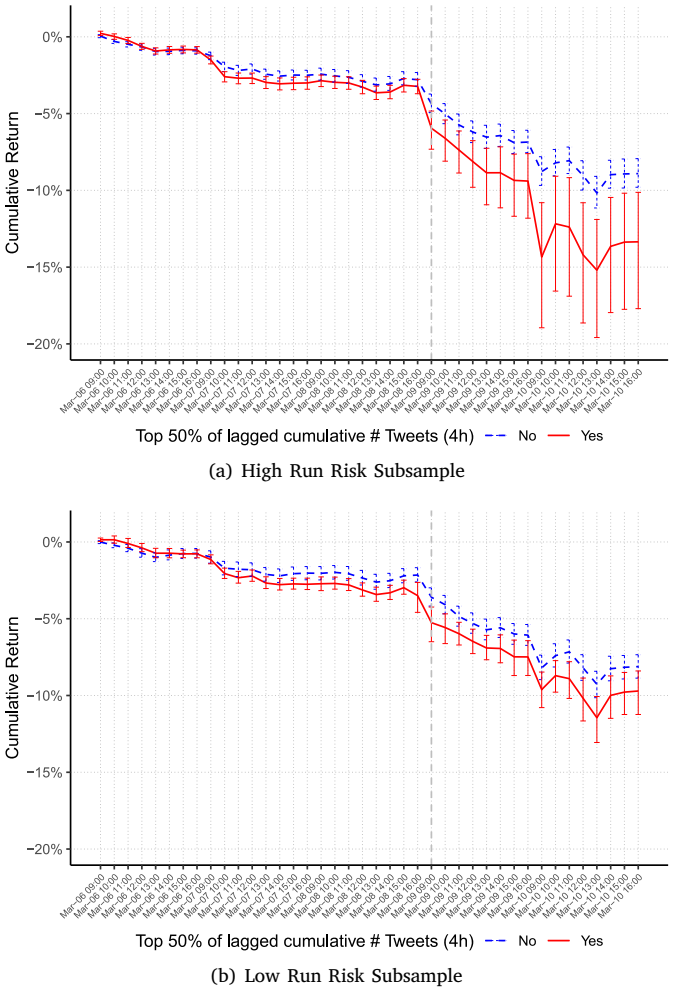
[20] The results remain similar when we use the continuous number of Tweets instead (see Appendix Table A.10).

four-hour window, and their interactions with the run indicator '$1(\geq$ Mar 09$)_t$'. We cluster standard errors by bank to account for arbitrary correlations in errors for observations within bank, alleviating concerns about standard errors due to *Run Risk*'s time-invariant nature (Petersen, 2008; Thompson, 2011).[21]

Panel A of Table 6 presents the results from estimating Eq. (5). Columns 1 and 4 present estimates from the specifications without fixed effects, and columns 2-3 and 5-6 add bank- and day-by-hour fixed effects. Across all specifications (columns 1–2 and 4–5) we estimate a large negative coefficient on '$1(\geq$ Mar 09$)_t$', indicating that bank stocks dropped significantly after the onset of the run across both subsamples. Next, focusing on banks with low run risk (i.e., a below-median product of *% Loss MTM* and *% Uninsured*) in columns 1 through 3, we estimate the interaction of '$1(\geq$ Mar 09$)_t$' and '1(N Tweets High)$_{i,t-1}$' to be small and insignificant across all three specifications. In contrast, in the high run risk sample (columns 4-6), we estimate a significant and negative interaction between '$1(\geq$ Mar 09$)_t$' and '1(N Tweets High)$_{i,t-1}$.' The magnitude of this interaction term is stable upon including fixed effects, indicating that after the run began for high run risk banks, returns decline by 0.1 bps more for banks with high recent Twitter attention than banks with low recent Twitter attention.[22] Put differently, we can attribute significantly lower hourly returns of stocks with high balance sheet risk to high *recent* Twitter attention.

We perform several robustness exercises on this result.[23] First, in Panel B of Table 6, we consider a shorter time window around the onset of the bank run period: one day before (March 8) through the first full day (March 9). We obtain coefficient estimates about twice as large over this shorter window (columns 1–3) for the subsample of high run risk banks compared to the longer window (columns 4–6), indicating that our findings are indeed driven by the period around the onset of the SVB run. Second, in Panel C of Table 6, we report the results when dropping Silicon Valley Bank (SIVB) from the sample, both for the shorter- and the longer event window. The interaction term remains statistically significant while being slightly smaller in magnitude, especially for the longer estimation window. Next, we repeat the specification using retweet-weighted tweet counts (i.e., counting 1 plus the number of retweets for each tweet). The results are nearly identical to using unweighted tweet counts (see Panel D of Table 6).[24] Further, we also exclude too-big-to-fail banks, finding similar results (Appendix Table A.11a).

To examine the price dynamics during the run period and to rule out pre-trends, we plot cumulative returns following high versus low Twitter attention in Fig. 8, separately for banks above versus below the median of run risk. For each trading hour between March 6 and March 10, we divide each set of banks into those above versus below the median of lagged (4-hour) Twitter attention, computing the average return. Separately for above-median and below-median Twitter attention, we sum these hourly average returns over the event window. To approximate standard errors, we perform a block bootstrap, sampling with replacement by bank for 1000 replications, which accounts for the accumulation of returns over the window and clusters errors by bank (Cameron et al., 2008). Consistent with the run period driving the interaction between run risk and Twitter attention, cumulative returns diverge meaningfully for the first time at the run's onset (March 9th), but returns do not diverge at all in the low run risk subsample.



(a) High Run Risk Subsample

(b) Low Run Risk Subsample

Fig. 8. *Cumulative Returns Following High versus Low Twitter Attention.* This figure shows the average cumulative returns of publicly listed banks and bank holding companies (BHCs) from March 06 until March 10, 2023 (during trading hours), splitting the sample into banks with high run risk (Panel (a)) and low run risk (Panel (b)). In each subsample we split the sample of bank stocks into those with above (red solid line) and below (blue dashed line) median number of tweets over the previous four hours, calculate average returns for the two groups, and cumulate the average returns at the hourly frequency from March 06 until March 10, 2023. The gray vertical dashed line indicates March 09, 2023 at 9am. 95% confidence intervals around the mean cumulative return are constructed by bootstrapping with clustering at the bank level.

These results suggest that social media played a role, not only in the bank run on SVB, but the banking distress that emerged more broadly. Next, we turn to examining heterogeneity, which provides more precise tests of social media's role, distinct from other channels of communication.

3.4. Heterogeneity in twitter attention

In this section, we consider two dimensions of heterogeneity. First, we consider alternative, shorter time lags to measure Twitter attention in the hourly tests. These tests draw a higher frequency connection

[21] We also consider double clustering by bank and date-time, which gives similar results (see Appendix Table A.11b).

[22] The interaction coefficient reflects how the average difference between bank-hours with high versus low Twitter attention in the past 4 h changes after the run began versus before the run. To quantify how the returns change during the run period for a bank with persistently high Twitter attention throughout the sample period, it is appropriate to add the coefficient on '1(N Tweets High)$_{i,t-1}$' to this interaction, which owing to the small main effect coefficients, gives similar magnitudes of −0.105 (column 4), −0.075 (column 5) and −0.069 (column 6).

[23] For brevity, we only show the results for the high run risk subsample of banks in these robustness tests.

[24] In the Appendix Table A.11c, we count retweets instead of original tweets, finding a similar result to the retweet-weighted result.

Table 6
Social media and the bank run — hourly frequency.

Panel A: Run risk and Tweet volume over the prior 4 h						
	Hourly stock return					
	Low run risk			High Run Risk		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.201***	−0.209***		−0.150***	−0.165***	
	(0.019)	(0.020)		(0.051)	(0.046)	
1(# Tweets (4 h) High) (t − 1)	−0.019**	0.007	0.007	−0.009	0.033**	0.031***
	(0.009)	(0.012)	(0.010)	(0.006)	(0.013)	(0.011)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	0.007	0.017	0.016	−0.096**	−0.108**	−0.100**
	(0.029)	(0.029)	(0.027)	(0.038)	(0.047)	(0.042)
Ret (Cum.-4 h) (z) (t − 1)	0.052	0.032	0.023	0.096*	0.042	0.028
	(0.036)	(0.036)	(0.046)	(0.056)	(0.044)	(0.068)
1(≥ Mar 09) × Ret (Cum.-4 h) (z) (t − 1)	−0.107***	−0.095***	−0.107***	−0.003	0.011	0.031
	(0.038)	(0.034)	(0.031)	(0.073)	(0.077)	(0.075)
N Articles (4 h) (z) (t − 1)	−0.001*	−0.003***	−0.001**	0.029**	0.221	0.146
	(0.000)	(0.000)	(0.000)	(0.013)	(0.148)	(0.147)
1(≥ Mar 09) × N Articles (4 h) (z) (t − 1)	0.006***	0.005***	0.007***	−0.176**	−0.014	−0.031
	(0.001)	(0.001)	(0.002)	(0.083)	(0.121)	(0.099)
Constant	−0.030***			−0.038***		
	(0.004)			(0.004)		
Observations	15,330	15,330	15,330	14,996	14,996	14,996
R^2	0.013	0.019	0.236	0.022	0.040	0.245
Within R^2		0.013	0.004		0.016	0.003
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

Panel B: Short- vs. long event window						
	Hourly stock return					
	March 08–09			March 06–10		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.589***	−0.677***		−0.150***	−0.165***	
	(0.169)	(0.162)		(0.051)	(0.046)	
1(# Tweets (4 h) High) (t − 1)	−0.008	0.036***	0.025**	−0.009	0.033**	0.031***
	(0.009)	(0.013)	(0.011)	(0.006)	(0.013)	(0.011)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	−0.186**	−0.214**	−0.147***	−0.096**	−0.108**	−0.100**
	(0.075)	(0.085)	(0.054)	(0.038)	(0.047)	(0.042)
Constant	−0.021***			−0.038***		
	(0.007)			(0.004)		
Observations	5577	5577	5577	14,996	14,996	14,996
R^2	0.162	0.213	0.359	0.022	0.040	0.245
Within R^2		0.155	0.053		0.016	0.003
Return Controls + Interactions	✓	✓	✓	✓	✓	✓
News Controls + Interactions	✓	✓	✓	✓	✓	✓
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

Panel C: Excluding SIVB from the sample						
	Hourly stock return					
	Short Window, Excl. SIVB			Long Window, Excl. SIVB		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.777***	−0.842***		−0.199***	−0.216***	
	(0.078)	(0.080)		(0.021)	(0.022)	
1(# Tweets (4 h) High) (t − 1)	−0.003	0.028*	0.020	−0.007	0.028**	0.028**
	(0.008)	(0.016)	(0.013)	(0.006)	(0.013)	(0.011)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	−0.184*	−0.203*	−0.152**	−0.091**	−0.081*	−0.079*
	(0.098)	(0.107)	(0.076)	(0.043)	(0.045)	(0.043)
Constant	−0.001			−0.036***		
	(0.007)			(0.004)		
Observations	5538	5538	5538	14,891	14,891	14,891
R^2	0.192	0.226	0.379	0.016	0.024	0.269
Within R^2		0.210	0.008		0.018	0.013
Return Controls + Interactions	✓	✓	✓	✓	✓	✓
News Controls + Interactions	✓	✓	✓	✓	✓	✓
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

(*continued on next page*)

Table 6 (*continued*).

Panel D: Retweet (RT)-weighted Tweets			
	Hourly stock return		
	High run risk		
	(1)	(2)	(3)
1(≥ Mar 09)	−0.160***	−0.176***	
	(0.055)	(0.050)	
1(# RT-weighted Tweets (4 h) High) (t − 1)	−0.008	0.035**	0.034***
	(0.006)	(0.015)	(0.012)
1(≥ Mar 09) × 1(# RT-weighted Tweets (4 h) High) (t − 1)	−0.078**	−0.090**	−0.082**
	(0.033)	(0.043)	(0.038)
Ret (Cum.-4 h) (z) (t − 1)	0.098*	0.043	0.030
	(0.056)	(0.044)	(0.068)
1(≥ Mar 09) × Ret (Cum.-4 h) (z) (t − 1)	−0.004	0.010	0.029
	(0.074)	(0.077)	(0.076)
N Articles (4 h) (z) (t − 1)	0.030**	0.223	0.147
	(0.013)	(0.148)	(0.147)
1(≥ Mar 09) × N Articles (4 h) (z) (t − 1)	−0.179**	−0.018	−0.036
	(0.084)	(0.122)	(0.100)
Constant	−0.039***		
	(0.004)		
Observations	14,786	14,786	14,786
R^2	0.022	0.040	0.246
Within R^2		0.016	0.003
Firm FE		✓	✓
Day-by-Hour FE			✓

This table presents OLS regression estimates that relate hourly bank stock returns to lagged Twitter attention, interacted with an indicator for the onset of the SVB run on March 9. The dependent variable in all columns is the hourly return (in basis points) for a bank stock in our sample. The sample is organized at the firm-by-day-by-hour level. We split the sample into below-median (columns 1–3) and above-media (columns 4–6) banks with respect to "Run Risk (z)", which is defined as '% Uninsured' × % Loss MTM' and estimate the regression separately for the two subsamples. The indicator variable '1(≥ Mar 9)' equals one after the onset of the SVB bank run on March 9, 2023, at 9am, and zero otherwise. Firm- and Day-by-Hour fixed effects are included as indicated. Lagged Twitter attention is an indicator variable denoted as '1(# Tweets (4 h) High) (t − 1),' which takes the value of 1 if the number of Tweets posted about the bank in the 4 hours prior to the current period t (i.e., hours $t-4$ through $t-1$) was above the median, and zero otherwise. We additionally control for the number of news articles posted about a bank, the bank's cumulative stock return in hours $t-5$ through $t-1$, and the interactions between these variables with the run period indicator in all columns. Panel A presents the baseline estimates. Panel B employs a shorter sample around from March 08 (9am) to March 09 (4pm) in columns 1–3. Panel C excludes Silicon Valley Bank (SIVB) from the sample, and Panel D uses the number of tweets weighted by the corresponding number of Retweets (within the next hour) to measure social media activity. Standard errors reported in parentheses are clustered at the bank level.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

between Twitter attention and returns. Second, we point to the importance of the Twitter community for amplifying the importance of Tweet activity, by comparing tweets which received a high- and low number of retweets. Table 7 presents the results of these exercises.

Panel A of Table 7 presents the results from estimating Eq. (5), but considering lagged Twitter attention in the past one hour and in the past two hours. Consistent with run period returns quickly reflecting Twitter attention, we estimate a similarly significant interaction term of '$1(\geq \text{Mar } 09)_t$' and '$1(\text{N Tweets High})_{i,t-1}$' when we use one-hour (column 1) and two-hour (column 2) lagged Twitter attention compared to Twitter activity over the previous four hours (column 3).

Next, in Panel B of Table 7, we consider heterogeneity in the social dimension of Twitter attention. Specifically, within the subsample of high run-risk banks, we split the observations further on whether the tweets from the previous 4 h received an above-median or below-median number of retweets, and separately estimate Eq. (5) for each subsample. We find a larger, more precisely estimated negative coefficient estimate for the subsample of high-retweets compared to low-retweets. For example, for the specification with all fixed effects in columns 3 and 6, the estimated coefficient is approximately 65% larger in the high-retweet subsample.

This finding points to the importance of social propagation of tweets in amplifying bank run risk during this episode. Beyond providing evidence of communication, this finding also helps to uniquely point to a social media mechanism versus some factor correlated with Twitter attention. It is not just that Twitter attention correlates with bank run risk, but the number of tweets that are retweeted matters.

Finally, we also consider a test that examines the amplification role of Twitter sentiment, distinct from Twitter attention. In an investment context, Cookson, Lu, Mullins, and Niessner (2024a) show that social media sentiment and attention predict returns in different directions. In this test, we similarly consider whether Twitter sentiment and attention have distinct implications for bank run risk. To do this, we compute the average VADER sentiment (Hutto and Gilbert, 2014) in a trailing 4-hour period for each bank-hour and estimate Eq. (5) while including Twitter sentiment along with Twitter attention. Appendix Table A.12 reports the results, which show small and statistically insignificant interactions between Twitter sentiment (VADER) and the run risk indicator. That is, lagged Twitter attention exhibits a significant amplification of bank run risks, but Twitter sentiment does not.

3.5. Tweet-level tests of market impacts of tweet sentiment

In this section, we investigate whether Twitter sentiment amplifies run risks at a higher frequency than the hourly tests. The empirical tests in this section analyze price changes in bank stocks over 5-minute intervals around run period tweets about banks, adapting the approach introduced in recent work on the impact of legislator and Presidential tweets (Bianchi et al., 2023, 2024). In our implementation of these high frequency tests, we study changes in log prices of bank stocks in 5-minute windows around the tweet timestamp as detailed in Section 2.3. This high frequency analysis allows for clean identification of the *immediate* impact of social media sentiment on stock prices. The narrow time window limits the scope for external factors — like news or price movements — to confound the observed relation between social

Table 7
Hourly frequency — different lags and tweet attention.

Panel A: Baseline specification with different lags			
	Hourly stock return		
	1 h	2 h	4 h
	(1)	(2)	(3)
1(# Tweets High) (t − 1)	0.038**	0.028*	0.031***
	(0.018)	(0.015)	(0.011)
1(≥ Mar 09) × 1(# Tweets High) (t − 1)	−0.074*	−0.084*	−0.100**
	(0.045)	(0.046)	(0.042)
Observations	14,996	14,996	14,996
R^2	0.244	0.244	0.245
Within R^2	0.003	0.003	0.003
Return Controls + Interactions	✓	✓	✓
News Controls + Interactions	✓	✓	✓
Firm FE	✓	✓	✓
Day-by-Hour FE	✓	✓	✓

Panel B: Tweets with high- vs low number of Retweets						
	Hourly stock return					
	Retweets low			Retweets High		
	(1)	(2)	(3)	(4)	(5)	(6)
1(≥ Mar 09)	−0.226***	−0.228***		−0.133**	−0.164***	
	(0.043)	(0.046)		(0.066)	(0.062)	
1(# Tweets (4 h) High) (t − 1)	0.005	0.047***	0.021*	−0.019**	0.006	0.035**
	(0.011)	(0.015)	(0.013)	(0.009)	(0.019)	(0.014)
1(≥ Mar 09) × 1(# Tweets (4 h) High) (t − 1)	−0.037	−0.047	−0.075*	−0.141***	−0.179***	−0.124**
	(0.051)	(0.065)	(0.039)	(0.053)	(0.068)	(0.062)
Constant	−0.036***			−0.033***		
	(0.006)			(0.006)		
Observations	7485	7485	7485	7511	7511	7,511
R^2	0.037	0.052	0.391	0.021	0.056	0.225
Within R^2		0.035	0.024		0.016	0.003
Return Controls + Interactions	✓	✓	✓	✓	✓	✓
News Controls + Interactions	✓	✓	✓	✓	✓	✓
Firm FE		✓	✓		✓	✓
Day-by-Hour FE			✓			✓

This table presents OLS regressions that relate hourly stock return (in bps) to lagged in-run Twitter attention measures, analogous to Table 6. We include only the subset of banks in the high-run risk subsample. Panel A presents specifications similar to these but considering tweet counts over the previous 1 h, 2 h, and 4 h, respectively, to construct '1(# Tweets High) (t − 1)' across columns 1–3. Panel B further splits the sample into observations with a high (above-median) and low (below-median) number of Retweets (RTs), and estimates the model separately for the two subsamples. Stock return and newspaper controls and their interactions with '1(≥ Mar 09)' as well as bank and day-by-hour fixed effects are included as indicated. Standard errors reported in parentheses are clustered at the bank level.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

media sentiment and changes in prices. However, since these tests are conducted at the tweet-level (one tweet is one observation), these tests are silent on how Twitter *attention* relates to high frequency returns.

We define the return of bank i's stock over the 5-minute window around a tweet authored at time t about bank i as the difference in log prices $\Delta p_{it} = p_{i,t+5'} - p_{i,t-\epsilon}$. In this formulation, $p_{i,t-\epsilon}$ and $p_{i,t+5'}$ denote the log price of the last trade *just* before the tweet and the log price of the first trade five minutes or more after the tweet.[25] This design allows five minutes for the price to adjust to the tweet. Since this high frequency return is fully realized in the five minutes after the tweet, the timing of the return alleviates concerns that stock prices drive Twitter activity, rather than the reverse. However, one might be concerned that recent price movements or news coverage (e.g., the 10 min prior to tweet) jointly drive the decision to tweet and the high frequency return. Thus, we also control for recent price changes and recent news coverage (both the number of news articles and their ESS-sentiment scores obtained from RavenPack), both in the 10 min before the tweet.

Specifically, we regress log price changes for bank i on the tone of the tweet and controls in a tweet-level specification:

$$\Delta p_{i,t} = a + b \times \text{VADER}Pos_{i,t} + c \times \text{VADER}Neg_{i,t} + \eta \mathbf{X}_{i,t} + \gamma_i + \epsilon_{i,t}, \qquad (6)$$

where $\Delta p_{i,t}$ is winsorized at the 1% level and expressed in basis points, $\text{VADER}Pos_{i,t}$ and $\text{VADER}Neg_{i,t}$ are the positive and negative components of the tweet's sentiment score assigned by VADER, and γ_i denotes bank fixed effects. To ease comparability, both negative and positive components have been standardized to have a mean of zero and a standard deviation of one in the full sample. In addition, the vector of controls $\mathbf{X}_i$ includes the log price change from 10 min prior to the tweet to just before the tweet, the number of news articles in the 10-minute window before the tweet according to Ravenpack, and the average ESS sentiment of those traditional news articles according to Ravenpack. We estimate specifications with and without these controls to show sensitivity to news and price changes. In all regressions, we cluster standard errors at the bank-day level, following Bianchi et al. (2024).

[25] We allow up to 10 min of staleness for each of these prices. The price *just before* the tweet is the last trade in the interval $[t - 10', t)$ while the price *five minutes after the tweet* is the first trade in the interval $[t + 5', t + 15']$.

Table 8
High frequency return responses to tweet sentiment.

	(1) $\Delta p_{i,t}$	(2) $\Delta p_{i,t}$	(3) $\Delta p_{i,t}$	(4) $\Delta p_{i,t}$	(5) $\Delta p_{i,t}$	(6) $\Delta p_{i,t}$	(7) $\Delta p_{i,t}$	(8) $\Delta p_{i,t}$	(9) $\Delta p_{i,t}$	(10) $\Delta p_{i,t}$
VADER Pos(z)	−1.83*	−1.70	−1.39	−1.31	0.76	1.05	−1.82*	−1.75	0.77	1.14
	(1.07)	(1.22)	(1.25)	(1.40)	(1.05)	(1.17)	(1.04)	(1.20)	(0.94)	(1.07)
VADER Neg(z)	2.64	3.04	2.93	3.34*	−0.18	0.18	3.07	3.49*	1.65	1.97
	(2.01)	(1.85)	(2.04)	(1.95)	(1.42)	(1.41)	(2.19)	(2.02)	(1.56)	(1.55)
Tech Community Flag	13.35*	12.64							13.34*	12.70
	(7.86)	(7.81)							(7.88)	(7.81)
VADER Pos(z) × Tech Community Flag	3.59	1.67							4.10	2.25
	(7.36)	(7.55)							(7.12)	(7.24)
VADER Neg(z) × Tech Community Flag	−12.03***	−11.14**							−11.34**	−10.44*
	(4.56)	(5.45)							(4.79)	(5.70)
Run Tweet			−1.31	−1.38					−2.22	−2.21
			(4.40)	(4.56)					(4.60)	(4.69)
VADER Pos(z) × Run Tweet			−5.42	−6.08					−6.14	−6.64
			(9.04)	(10.17)					(9.10)	(10.16)
VADER Neg(z) × Run Tweet			−13.27***	−12.91***					−12.73***	−12.35***
			(2.93)	(2.81)					(2.69)	(2.76)
VADER Pos(z) × High Run Risk (M2)					−2.97*	−3.26*			−3.05*	−3.31*
					(1.59)	(1.81)			(1.62)	(1.83)
VADER Neg(z) × High Run Risk (M2)					2.99	3.07			3.08	3.15
					(2.72)	(2.57)			(2.87)	(2.72)
Contagion Tweet							15.37	11.27	16.44	12.34
							(26.32)	(25.70)	(26.54)	(26.01)
VADER Pos(z) × Contagion Tweet							18.99	17.33	18.55	17.02
							(12.36)	(12.08)	(12.54)	(12.20)
VADER Neg(z) × Contagion Tweet							−22.36**	−20.63*	−22.63**	−20.91*
							(11.00)	(11.19)	(11.14)	(11.35)
$p_{i,t-\epsilon} - p_{i,t-10}$		0.08**		0.08**		0.08**		0.08**		0.08**
		(0.03)		(0.03)		(0.03)		(0.03)		(0.03)
Total News (−10, −ϵ) (z)		0.77		0.70		0.75		0.82		0.78
		(3.58)		(3.67)		(3.59)		(3.61)		(3.63)
ESS (−10, −ϵ) (z)		17.92***		17.90***		17.93***		17.98***		17.93***
		(4.15)		(4.19)		(4.14)		(4.12)		(4.16)
Constant	−15.29***	−11.59***	−14.57***	−10.89***	−14.82***	−11.15***	−14.53***	−10.84***	−14.80***	−11.08***
	(4.04)	(2.42)	(4.03)	(2.37)	(3.84)	(2.28)	(3.92)	(2.42)	(4.18)	(2.55)
Observations	36751	36659	36751	36659	36751	36659	36751	36659	36751	36659
R^2 (%)	1.07	2.5	1.06	2.5	1.05	2.49	1.08	2.51	1.11	2.54
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09

This table presents OLS estimates for how high frequency returns from *just* before the tweet to 5 minutes after relate to tweet sentiment interacted with whether the tweet is run-relevant (“run” or “contagion” tweets, or authored by tech community users) and whether the bank has high run risk. The level of observation is at the individual tweet level. In all regressions, the dependent variable $\Delta p_{i,t}$ is the log change in bank i's price between the last trade before the Tweet was posted and the first trade in the $[+5, +15]$ minute window around the tweet at time t. $\Delta p_{i,t}$ is winsorized at the 1% level and expressed in basis points. VADER Pos (z) and VADER Neg (z) are the positive and negative components of the VADER sentiment respectively standardized using the sample used in each regression. ‘Tech Community Flag’ indicates that a given tweet was posted by a member of the tech community, as constructed in Section Section 2.1.2. ‘Contagion Tweet’ and ‘Run Tweet’ indicate if the tweet contains at least one token in the contagion and run behavior dictionaries respectively. ‘High Run Risk’ indicates if the bank is in the top quintile of bank run risk (i.e., MTM losses × uninsured deposits). We exclude observations for which the traded volume associated with the last price in the window before the tweet or the first price in the window after the tweet is zero. In all regressions, we include bank fixed effects. Specifications in even columns additionally control for the bank stock return in the 10 minutes prior to the tweet (‘$p_{i,t-\epsilon} - p_{i,t-10}$’) We also control for the number of articles posted in the 10 minutes before the Tweet (‘Total News’) and the corresponding news sentiment (‘ESS’) using data from RavenPack. Standard errors are reported in parenthesis and clustered at the bank-day level.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

Our primary interest is in evaluating how prices respond to tweet sentiment for different, influential sets of tweets, as well as across banks with high versus low balance sheet risk.[26] Our main specification interacts sentiment (VADER *Pos* and VADER *Neg*) with indicators for whether the tweet is written by a ‘tech community’ user, is classified as a ‘run’ tweet or a ‘contagion’ tweet, or refers to a bank in the top quintile of bank run risk. Of particular interest, the coefficient on the interaction between VADER *Neg* and the *High Run Risk* indicator tests whether negative tweet sentiment amplifies bank run risks at high frequency.

Table 8 reports the results from estimating Eq. (6), interacting sentiment with tweet type. Odd columns are estimated without controls while even columns include controls for recent price changes, news coverage, and news sentiment. The estimates on VADER *Neg*(z) × *Tweet Type* indicate that negative sentiment tweets by tech community users and tweets about running and contagion correspond to large, immediate and negative price responses. Comparing the odd and even columns, we obtain very similar estimated magnitudes whether or not we control for recent news or price movements.

Turning to the amplification hypothesis, the estimated interaction coefficient between *High Run Risk* and VADER *Neg* (z) is small and statistically insignificant, irrespective of controlling for other factors. Thus, as in the hourly frequency tests, there is no evidence that Twitter *sentiment* amplifies bank run risks, even at high frequency. This finding on Twitter sentiment contrasts with our robust evidence that Twitter

[26] Returns are more responsive to tweet sentiment during the run period than they are outside of the run period. We show this in Appendix Table A.13, which presents estimates for both the full sample and the run period.

Table 9
High frequency tests, Controlling for retweets and followers.

	(1) $\Delta p_{i,t}$	(2) $\Delta p_{i,t}$	(3) $\Delta p_{i,t}$	(4) $\Delta p_{i,t}$	(5) $\Delta p_{i,t}$	(6) $\Delta p_{i,t}$	(7) $\Delta p_{i,t}$	(8) $\Delta p_{i,t}$
VADER Pos(z)	−1.70	−2.49	−1.31	−2.13	−1.75	−2.56	1.05	0.03
	(1.22)	(1.76)	(1.40)	(1.99)	(1.20)	(1.79)	(1.17)	(1.53)
VADER Neg(z)	3.04	2.71	3.34*	3.13*	3.49*	3.07*	0.18	0.06
	(1.85)	(1.79)	(1.95)	(1.88)	(2.02)	(1.84)	(1.41)	(1.89)
Tech Community Flag	12.64	13.55*						
	(7.81)	(7.68)						
VADER Pos(z) × Tech Community Flag	1.67	0.68						
	(7.55)	(7.87)						
VADER Neg(z) × Tech Community Flag	−11.14**	−11.38**						
	(5.45)	(5.53)						
Run Tweet			−1.38	−0.20				
			(4.56)	(4.08)				
VADER Pos(z) × Run Tweet			−6.08	−8.10				
			(10.17)	(10.86)				
VADER Neg(z) × Run Tweet			−12.91***	−13.81***				
			(2.81)	(3.06)				
Contagion Tweet					11.27	11.41		
					(25.70)	(25.90)		
VADER Pos(z) × Contagion Tweet					17.33	16.59		
					(12.08)	(11.67)		
VADER Neg(z) × Contagion Tweet					−20.63*	−20.62*		
					(11.19)	(11.10)		
VADER Pos(z) × High Run Risk (M2)							−3.26*	−3.02*
							(1.81)	(1.78)
VADER Neg(z) × High Run Risk (M2)							3.07	2.84
							(2.57)	(2.44)
High Followers (M2)		−1.25		−0.94		−0.99		−0.89
		(3.72)		(3.86)		(3.74)		(3.70)
VADER Pos(z) × High Followers (M2)		0.44		0.52		0.43		0.44
		(3.17)		(3.10)		(3.11)		(3.17)
VADER Neg(z) × High Followers (M2)		1.22		0.95		1.38		1.13
		(2.84)		(2.76)		(2.98)		(2.80)
High Retweets (M2)		−10.67		−10.43		−9.99		−10.02
		(8.56)		(8.07)		(8.50)		(8.49)
VADER Pos(z) × High Retweets (M2)		12.59***		14.15***		12.65***		12.70***
		(3.43)		(4.41)		(3.45)		(3.37)
VADER Neg(z) × High Retweets (M2)		−1.52		−1.26		−1.34		−1.68
		(6.43)		(6.67)		(6.73)		(6.44)
$p_{i,t-\epsilon} - p_{i,t-10}$	0.08**	0.08**	0.08**	0.08**	0.08**	0.08**	0.08**	0.08**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Total News (−10, $-\epsilon$) (z)	0.77	0.78	0.70	0.69	0.82	0.83	0.75	0.76
	(3.58)	(3.62)	(3.67)	(3.70)	(3.61)	(3.64)	(3.59)	(3.62)
ESS (−10, $-\epsilon$) (z)	17.92***	17.95***	17.90***	17.90***	17.98***	18.00***	17.93***	17.95***
	(4.15)	(4.18)	(4.19)	(4.23)	(4.12)	(4.15)	(4.14)	(4.18)
Constant	−11.59***	−10.31***	−10.89***	−9.77***	−10.84***	−9.68***	−11.15***	−10.01***
	(2.42)	(3.25)	(2.37)	(3.37)	(2.42)	(3.27)	(2.28)	(3.16)
Observations	36 659	36 659	36 659	36 659	36 659	36 659	36 659	36 659
R^2 (%)	2.5	2.52	2.5	2.52	2.51	2.53	2.49	2.51
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09	≥ Mar09

This table presents OLS estimates for how high frequency returns relate to tweet sentiment interacted with whether the tweet is run-relevant ("run" or "contagion" tweets, or authored by tech community users) and whether the bank has high run risk, *controlling for measures of tweet influence*. The specifications and variable definitions are identical to Table 8, except that — in even columns — we additionally control for 'High Followers' and 'High Retweets,' which indicate if the author of the tweet is in the top quintile number of followers and if the Tweet received top quintile number of Retweets, respectively. Standard errors are reported in parenthesis and doubled clustered at the bank-day level.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

attention amplifies classical run risk factors. Taken together, it matters more *who* is communicating (e.g., tech community users who are likely depositors) and whether their communication is run relevant than whether someone unconnected to the run tweets with negative sentiment (e.g., investors observing the event).

Next, we estimate a version of Eq. (6) that controls for whether the tweet is in the top quintile of retweets (*High Retweets*) or the author is in the top quintile of followers (*High Followers*). These tests address the concern that the price impacts of tech community tweets are primarily due to these users having more followers, and thus, tweets with greater reach. Table 9 reports the results, which additionally control for the interactions of *High Followers* and *High Retweets* with VADER sentiment components. The inclusion of these terms does not change the estimated interactions between VADER *Neg* (z) and indicators for tech community tweets, run tweets and contagion tweets. Thus, the outsized price impact of these tweet types likely reflects the content of these tweets rather than underlying variation in their social influence.[27]

[27] Table A.14 presents our main high frequency tests on the sample that excludes tweets about Silicon Valley Bank (SIVB). On this much smaller sample, the main takeaways remain unchanged while the statistical significance is somewhat weaker (particularly on 'contagion' tweets). We continue to find that the negative tweet sentiment component has a significant negative effect on

Overall, these results emphasize the importance of run-relevant tweets (i.e., those authored by the tech community and those about 'running' and 'contagion') in driving the price dynamics during the SVB run period. Combined with evidence on the importance of Twitter *attention* to amplifying bank run risk, this finding reinforces our interpretation that Twitter attention and online conversation among likely depositors in the tech community were important for bank run risks.

4. Conclusion

This paper presents evidence that exposure to social media amplifies classical bank run risks. Our empirical tests show that banks with a large Twitter *pre-exposure* experienced a more severe run in March 2023 in the wake of SVB's collapse, particularly if they had large mark-to-market losses and a large percentage of uninsured deposits. At the hourly frequency, we show that negative returns emerge after periods of intense Twitter attention, but this effect only emerges after the run on SVB begins. Although we find strong evidence that Twitter *attention* predicts greater run severity for banks with greater run risks, negative Twitter *sentiment* does not.

The effect that we uncover — though a modern phenomenon linked to rapid communication over social media — is one that is rooted in classical bank run models, going back to Diamond and Dybvig (1983). Some of these models have posited that communication among depositors poses a risk to banks, especially when many of the deposits in the bank are uninsured (Kelly and Gráda, 2000). Increasingly, in today's society, social media provides a means for individuals to coordinate and communicate beyond what older technologies allow. The amplification of bank run risk via Twitter is a unique opportunity to observe communication and coordination that shapes a critically important economic outcome — distress in banks. Given the increasingly pervasive nature of social communication on and off Twitter, we do not expect this risk to go away, but rather, it is likely to influence outcomes in other domains, as well.

CRediT authorship contribution statement

J. Anthony Cookson: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Corbin Fox:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Javier Gil-Bazo:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Juan F. Imbet:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Christoph Schiller:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: J. Anthony Cookson reports a relationship with Cornerstone Research that includes: paid expert testimony. Given his role as AE of JFE, J. Anthony Cookson had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor." If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

high frequency bank stock returns, and this effect is much stronger when the tweet was posted by a member of the tech community or refers to 'running.'

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2025.104218.

Data availability

Social Media as a Bank Run Catalyst (Original data) (Mendeley Data)

References

Acemoglu, D., Ozdaglar, A., Tahbaz-Salehi, A., 2015. Systemic risk and stability in financial networks. Am. Econ. Rev. 105 (2), 564–608.

Acharya, V.V., Richardson, M.P., Schoenholtz, K.L., Tuckman, B., et al., 2023. SVB and beyond: The banking stress of 2023. NYU Stern White Paper.

Akçay, E., Hirshleifer, D., 2021. Social finance as cultural evolution, transmission bias, and market dynamics. Proc. Natl. Acad. Sci. 118 (26), e2015568118.

Angeletos, G.M., Werning, I., 2006. Crises and prices: Information aggregation, multiplicity, and volatility. Am. Econ. Rev. 96 (5), 1720–1736.

Baron, M., Schularick, M., Zimmermann, K., 2023. Survival of the biggest: Large banks and financial crises. Available at SSRN 4189014.

Begenau, J., Piazzesi, M., Schneider, M., 2015. Banks' Risk Exposures. Technical Report, National Bureau of Economic Research.

Bianchi, F., Gómez-Cram, R., Kind, T., Kung, H., 2023. Threats to central bank independence: High-frequency identification with twitter. J. Monet. Econ. 135, 37–54. http://dx.doi.org/10.1016/j.jmoneco.2023.01.001, URL: https://www.sciencedirect.com/science/article/pii/S0304393223000077.

Bianchi, F., Gomez-Cram, R., Kung, H., 2024. Using social media to identify the effects of congressional viewpoints on asset prices. Rev. Financ. Stud. 37, 2244–2272.

Brown, J.R., Cookson, J.A., Heimer, R.Z., 2019. Growing up without finance. J. Financ. Econ. 134 (3), 591–616.

Brunnermeier, M.K., 2009. Deciphering the liquidity and credit crunch 2007–2008. J. Econ. Perspect. 23 (1), 77–100.

Buchak, G., Matvos, G., Piskorski, T., Seru, A., 2018. Fintech, regulatory arbitrage, and the rise of shadow banks. J. Financ. Econ. 130 (3), 453–483.

Buchak, G., Matvos, G., Piskorski, T., Seru, A., 2024. Beyond the balance sheet model of banking: Implications for bank regulation and monetary policy. J. Political Econ. 132 (2), 616–693.

Caglio, C., Dlugosz, J., Rezende, M., 2023. Flight to safety in the regional bank crisis of 2023. Available at SSRN 4457140.

Calomiris, C.W., Kahn, C.M., 1991. The role of demandable debt in structuring optimal banking arrangements. Am. Econ. Rev. 81 (3), 497–513.

Calomiris, C.W., Mason, J.R., 1997. Contagion and bank failures during the great depression: The June 1932 chicago banking panic. Am. Econ. Rev. 87 (5), 863–883.

Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. Rev. Econ. Stat. 90 (3), 414–427.

Cassella, S., Dim, C., Karimli, T., 2023. Optimism shifting. Available at SSRN 4557313.

Chen, H., De, P., Hu, Y.J., Hwang, B.H., 2014. Wisdom of crowds: The value of stock opinions transmitted through social media. Rev. Financ. Stud. 27 (5), 1367–1403.

Chetty, R., et al., 2022a. Social capital I: measurement and associations with economic mobility. Nature 608 (7921), 108–121.

Chetty, R., et al., 2022b. Social capital II: determinants of economic connectedness. Nature 608 (7921), 122–134.

Choi, D.B., Goldsmith-Pinkham, P., Yorulmazer, T., 2023. Contagion Effects of the Silicon Valley Bank Run. Technical Report, National Bureau of Economic Research.

Chwe, M.S.Y., 2000. Communication and coordination in social networks. Rev. Econ. Stud. 67 (1), 1–16.

Cipriani, M., Eisenbach, T.M., Kovner, A., 2024. Tracing bank runs in real time. FRB of New York Staff Report, 1104.

Cookson, J.A., Engelberg, J.E., Mullins, W., 2020. Does partisanship shape investor beliefs? Evidence from the COVID-19 pandemic. Rev. Asset Pricing Stud. 10 (4), 863–893.

Cookson, J.A., Engelberg, J.E., Mullins, W., 2023. Echo chambers. Rev. Financ. Stud. 36 (2), 450–500. http://dx.doi.org/10.1093/rfs/hhac058, arXiv:https://academic.oup.com/rfs/advance-article-pdf/doi/10.1093/rfs/hhac058/45842263/hhac058.pdf. hhac058.

Cookson, J.A., Lu, R., Mullins, W., Niessner, M., 2024a. The social signal. J. Financ. Econ. 158 (8), 103870.

Cookson, J.A., Mullins, W., Niessner, M., 2024b. Social media and finance. Oxf. Res. Encycl. Econ. Financ. November.

Cookson, J.A., Niessner, M., 2020. Why don't we agree? Evidence from a social network of investors. J. Financ. 75 (1), 173–228.

Cookson, J.A., Niessner, M., Schiller, C., 2025. Can social media inform corporate decisions? Evidence from merger withdrawals. J. Financ. (Forthcoming).

Cramer, K.F., Koont, N., 2021. Peer effects in deposit markets. Available at SSRN 3930699.

Diamond, D.W., Dybvig, P.H., 1983. Bank runs, deposit insurance, and liquidity. J. Political Econ. 91 (3), 401–419.
Diamond, D.W., Rajan, R.G., 2000. A theory of bank capital. J. Financ. 55 (6), 2431–2465.
Dim, C., 2025. Social media analysts' skills: Insights from text-implied beliefs. J. Financ. Quant. Anal. 60 (7), 3081–3115.
Drechsler, I., Savov, A., Schnabl, P., 2017. The deposits channel of monetary policy. Q. J. Econ. 132 (4), 1819–1876.
Drechsler, I., Savov, A., Schnabl, P., 2021. Banking on deposits: Maturity transformation without interest rate risk. J. Financ. 76 (3), 1091–1143.
Drechsler, I., Savov, A., Schnabl, P., Wang, O., 2024. Deposit Franchise Runs. Technical Report, National Bureau of Economic Research.
Dupont, Q., 2022. Faith and finance: The role of trust in religious institutions on stock market participation. SSRN Working Paper 3875648.
Egan, M., Hortaçsu, A., Matvos, G., 2017. Deposit competition and financial fragility: Evidence from the us banking sector. Am. Econ. Rev. 107 (1), 169–216.
Elliott, M., Golub, B., Jackson, M.O., 2014. Financial networks and contagion. Am. Econ. Rev. 104 (10), 3115–3153.
Falato, A., Hortacsu, A., Li, D., Shin, C., 2021. Fire-sale spillovers in debt markets. J. Financ. 76 (6), 3055–3102.
Flannery, M.J., 1981. Market interest rates and commercial bank profitability: An empirical investigation. J. Financ. 36 (5), 1085–1101.
Flannery, M.J., James, C.M., 1984. Market evidence on the effective maturity of bank assets and liabilities. J. Money Credit. Bank. 16 (4), 435–445.
Fuster, A., Plosser, M., Schnabl, P., Vickery, J., 2019. The role of technology in mortgage lending. Rev. Financ. Stud. 32 (5), 1854–1899.
Gentzkow, M., Shapiro, J.M., 2011. Ideological segregation online and offline. Q. J. Econ. 126 (4), 1799–1839.
Giannini, R., Irvine, P., Shu, T., 2019. The convergence and divergence of investors' opinions around earnings news: Evidence from a social network. J. Financ. Mark. 42, 94–120.
Gil-Bazo, J., Imbet, J.F., 2025. Tweeting for money: Social media and mutual fund flows. Manag. Sci. (Forthcoming).
Goldsmith-Pinkham, P.S., Hirtle, B., Lucca, D.O., 2016. Parsing the content of bank supervision. FRB of NY Staff Report.
Goldstein, I., Pauzner, A., 2005. Demand–deposit contracts and the probability of bank runs. J. Financ. 60 (3), 1293–1327.
Golub, B., Jackson, M.O., 2010. Naive learning in social networks and the wisdom of crowds. Am. Econ. J.: Microecon. 2 (1), 112–149.
Gorton, G., Pennacchi, G., 1990. Financial intermediaries and liquidity creation. J. Financ. 45 (1), 49–71.
Granja, J., 2013. The relation between bank resolutions and information environment: Evidence from the auctions for failed banks. J. Account. Res. 51 (5), 1031–1070.
Granja, J., Jiang, E.X., Matvos, G., Piskorski, T., Seru, A., 2024. Book value risk management of banks: Limited hedging, HTM accounting and rising interest rates. Working Paper, SSRN Number 4410201.
Gruenberg, M., 2023. Remarks on oversight of financial regulators: Financial stability, supervision, and consumer protection in the wake of recent bank failures. URL: https://www.fdic.gov/news/speeches/2023/spmay1723.html. (Accessed 13 August 2024). Speech, Committee on Banking, Housing, and Urban Affairs, United States Senate.
Gurun, U.G., Stoffman, N., Yonker, S.E., 2018. Trust busting: The effect of fraud on investor behavior. Rev. Financ. Stud. 31 (4), 1341–1376.
Haddad, V., Hartman-Glaser, B., Muir, T., 2023. Bank fragility when depositors are the asset. Available at SSRN 4412256.
Haendler, C., 2022. Keeping up in the digital era: How mobile technology is reshaping the banking sector. Available at SSRN 4287985.
Han, B., Hirshleifer, D., Walden, J., 2022. Social transmission bias and investor behavior. J. Financ. Quant. Anal. 57 (1), 390–412.
Hirshleifer, D., 2020. Presidential address: Social transmission bias in economics and finance. J. Financ. 75 (4), 1779–1831.
Hirshleifer, D., Plotkin, J.B., 2021. Moonshots, investment booms, and selection bias in the transmission of cultural traits. Proc. Natl. Acad. Sci. 118 (26), e2015571118.
Hutto, C., Gilbert, E., 2014. Vader: A parsimonious rule-based model for sentiment analysis of social media text. In: Proceedings of the International AAAI Conference on Web and Social Media, vol. 8, no. 1, pp. 216–225.
Iyer, R., Puri, M., 2012. Understanding bank runs: The importance of depositor-bank relationships and networks. Am. Econ. Rev. 102 (4), 1414–1445.
Jiang, E.X., Matvos, G., Piskorski, T., Seru, A., 2024a. Monetary tightening and US bank fragility in 2023: Mark-to-market losses and uninsured depositor runs? J. Financ. Econ. 159, 103899.
Jiang, E., Matvos, G., Piskorski, T., Seru, A., 2024b. Which banks are (over) levered? Insights from shadow banks and uninsured leverage. NBER Working Paper.
Jiang, E.X., Matvos, G., Piskorski, T., Seru, A., 2025. U.S. bank fragility to credit risk in 2023: Monetary tightening and commercial real estate distress. J. Political Econ. - Macroecon. (Forthcoming).
Kearney, M.S., Levine, P.B., 2019. Early childhood education by television: Lessons from Sesame Street. Am. Econ. J.: Appl. Econ. 11 (1), 318–350.
Kelly, M., Gráda, C.Ó., 2000. Market contagion: Evidence from the panics of 1854 and 1857. Am. Econ. Rev. 90 (5), 1110–1124.
Koont, N., Santos, T., Zingales, L., 2023. Destabilizing digital 'bank walks'. Technical Report, New Working Paper Series.
Kuchler, T., Stroebel, J., 2021. Social finance. Annu. Rev. Financ. Econ. 13, 37–55.
Levy, R., 2021. Social media, news consumption, and polarization: Evidence from a field experiment. Am. Econ. Rev. 111 (3), 831–870.
Loughran, T., McDonald, B., 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. J. Financ. 66 (1), 35–65.
Loughran, T., McDonald, B., 2020. Textual analysis in finance. Annu. Rev. Financ. Econ. 12, 357–375.
Morris, S., Shin, H.S., 2000. Rethinking multiple equilibria in macroeconomic modeling. NBER Macroecon. Annu. 15, 139–161.
Müller, K., Pan, Y., Schwarz, C., 2023. Social media and stock market participation. Available at SSRN 4557783.
Müller, K., Schwarz, C., 2021. Fanning the flames of hate: Social media and hate crime. J. Eur. Econ. Assoc. 19 (4), 2131–2167.
Müller, K., Schwarz, C., 2023. From hashtag to hate crime: Twitter and anti-minority sentiment. Am. Econ. J. Appl. Econ. 15 (3), 270–312.
Parmelee, J.H., Bichard, S.L., 2011. Politics and the Twitter Revolution: How Tweets Influence the Relationship Between Political Leaders and the Public. Lexington books.
Peck, J., Shell, K., 2003. Equilibrium bank runs. J. Political Econ. 111 (1), 103–123.
Pedersen, L.H., 2022. Game on: Social networks and markets. J. Financ. Econ. 146 (3), 1097–1119.
Pennacchi, G., 2006. Deposit insurance, bank regulation, and financial system risks. J. Monet. Econ. 53 (1), 1–30.
Petersen, M.A., 2008. Estimating standard errors in finance panel data sets: Comparing approaches. Rev. Financ. Stud. 22 (1), 435–480.
Purnanandam, A., 2007. Interest rate derivatives at commercial banks: An empirical investigation. J. Monet. Econ. 54 (6), 1769–1808.
Renault, T., 2017. Intraday online investor sentiment and return patterns in the US stock market. J. Bank. Financ. 84, 25–40.
Simonsohn, U., Simmons, J.P., Nelson, L.D., 2020. Specification curve analysis. Nat. Hum. Behav. 4 (11), 1208–1214.
Stein, L.C., Yannelis, C., 2020. Financial inclusion, human capital, and wealth accumulation: Evidence from the Freedman's Savings Bank. Rev. Financ. Stud. 33 (11), 5333–5377.
Strömberg, D., 2004. Radio's impact on public spending. Q. J. Econ. 119 (1), 189–221.
Thompson, S.B., 2011. Simple formulas for standard errors that cluster by both firm and time. J. Financ. Econ. 99 (1), 1–10.
Traweek, V., Wardlaw, M., 2025. Societal inequality and financial market participation: Evidence from the Freedman's Savings Bank. Rev. Financ. Stud. (Forthcoming).
Van Bommel, J., 2003. Rumors. J. Financ. 58 (4), 1499–1520.
Vuillemey, G., 2023. From the saving glut to financial instability: Evidence from the Silicon Valley Bank failure. Available at SSRN 4413287.
Ziebarth, N.L., 2017. The Radio and Bank Distress in the Great Depression. University of Iowa, Manuscript.